



GIS in Fire and Rescue: Network Datasets and Roadway Routing

FAIRFAX COUNTY FIRE AND RESCUE DEPARTMENT

Chip Galloway & John Woolverton

Introduction

- Chip Galloway
- George Mason University BA in Geography 2008
- GIS Analyst since 2005
 - Federal contractor 2005 – 2006
 - Fairfax County Department of Transportation 2006 – 2008
 - Fairfax County Public Works (maintenance) 2008 – 2018
 - Fairfax County Public Works (planning) 2018 – 2021
 - Fairfax County Information Technology 2021 – 2024
 - Fairfax County Fire & Rescue Sept 2024



Introduction

- John Woolverton
- University of Mary Washington BA in Geography 2016
- GIS Analyst since 2017
 - Federal contractor 2017-2022 (Census Bureau – Geography Division)
 - Fairfax County Public Works (Directors Office - GIS) 2022 - 2025
 - Fairfax County Fire & Rescue April 2025



What is a Network Dataset?

A network dataset models the real-world transportation system as a connected, routable roadway network.

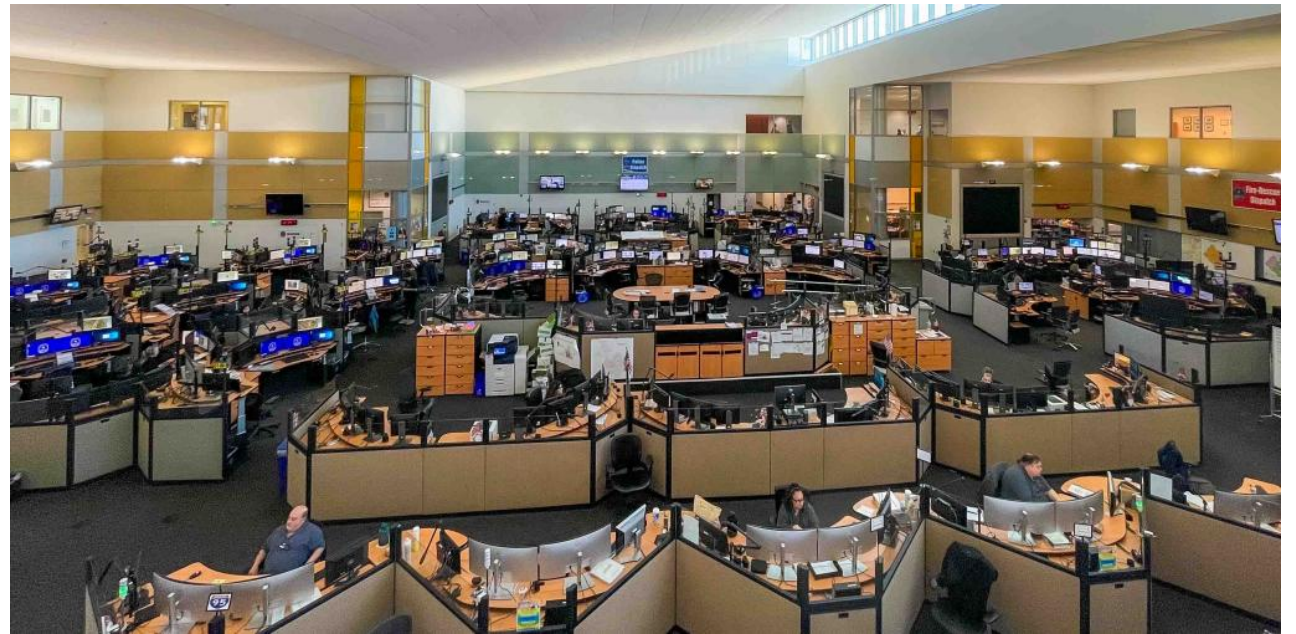
It represents streets, intersections, and travel rules so routing tools can perform navigation, calculate travel times, and determine optimal paths between locations.



Why do we maintain our own dataset?

Computer Aided Dispatch (CAD) uses the roadway centerline data to calculate and assign first responders to incidents.

- Supports CAD Operations
- Authoritative Data Source
- Local Control
- Operational Flexibility
- Controlled Map Updates

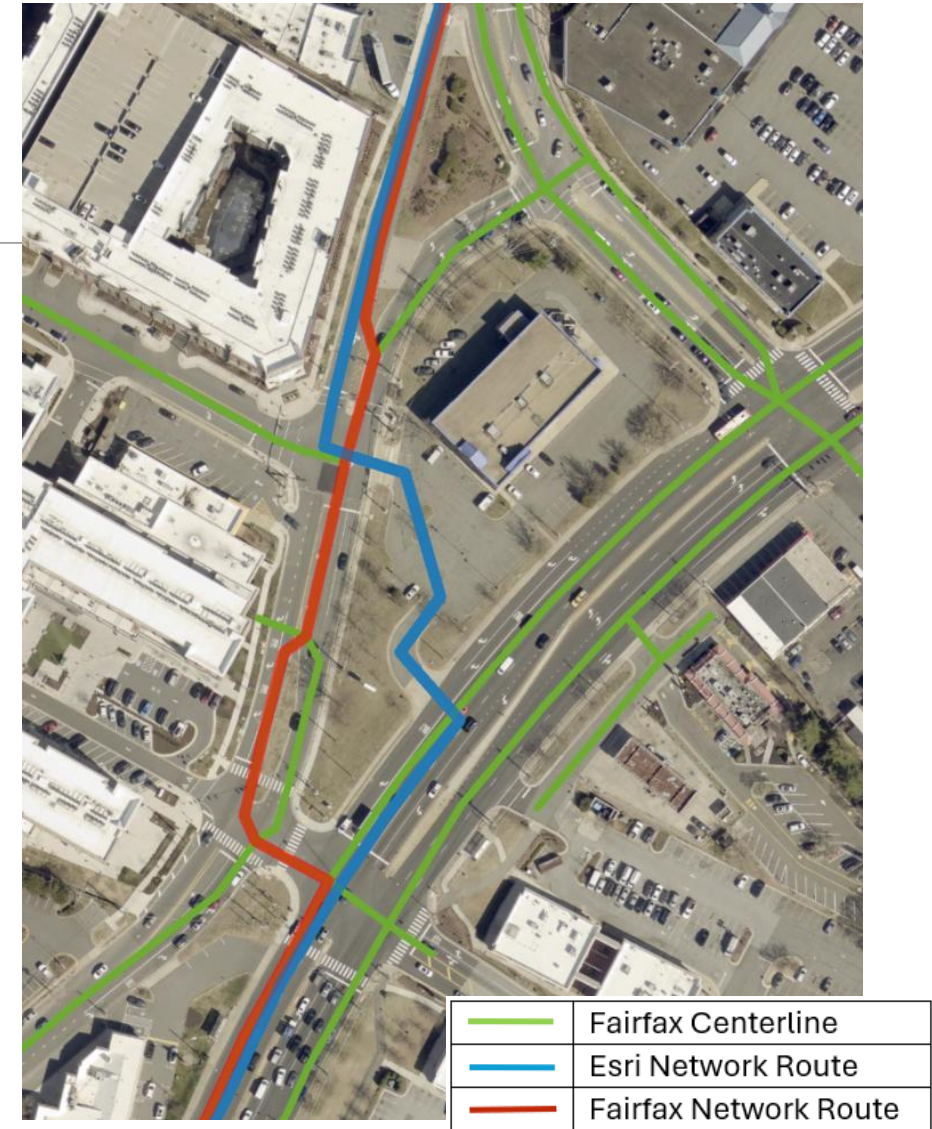


911 call takers and dispatchers

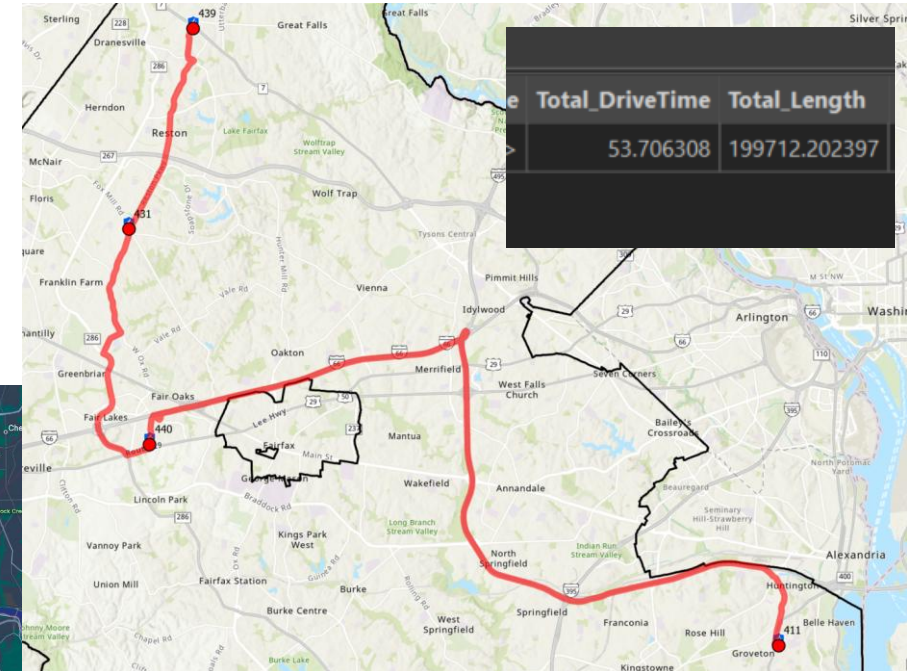
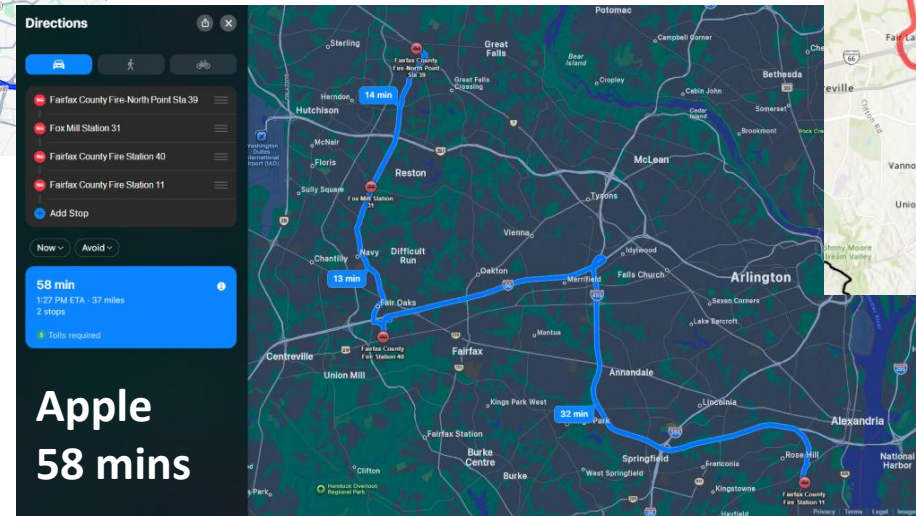
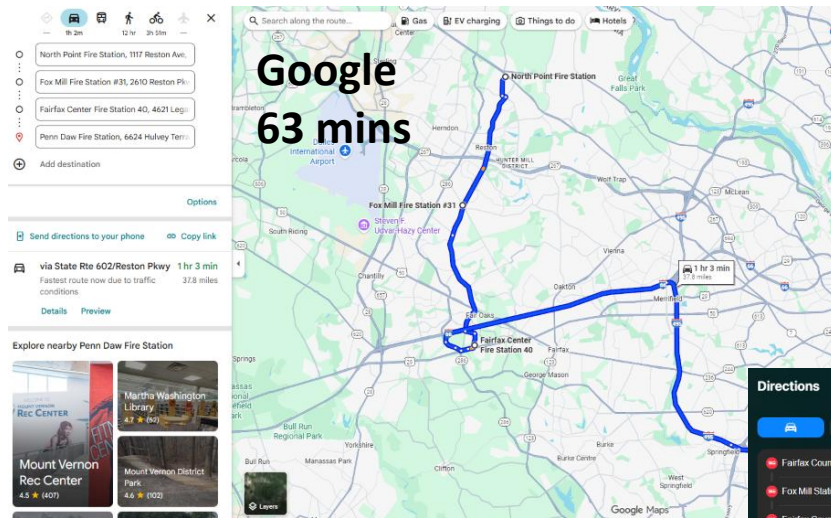
Why don't we use Esri's network dataset?

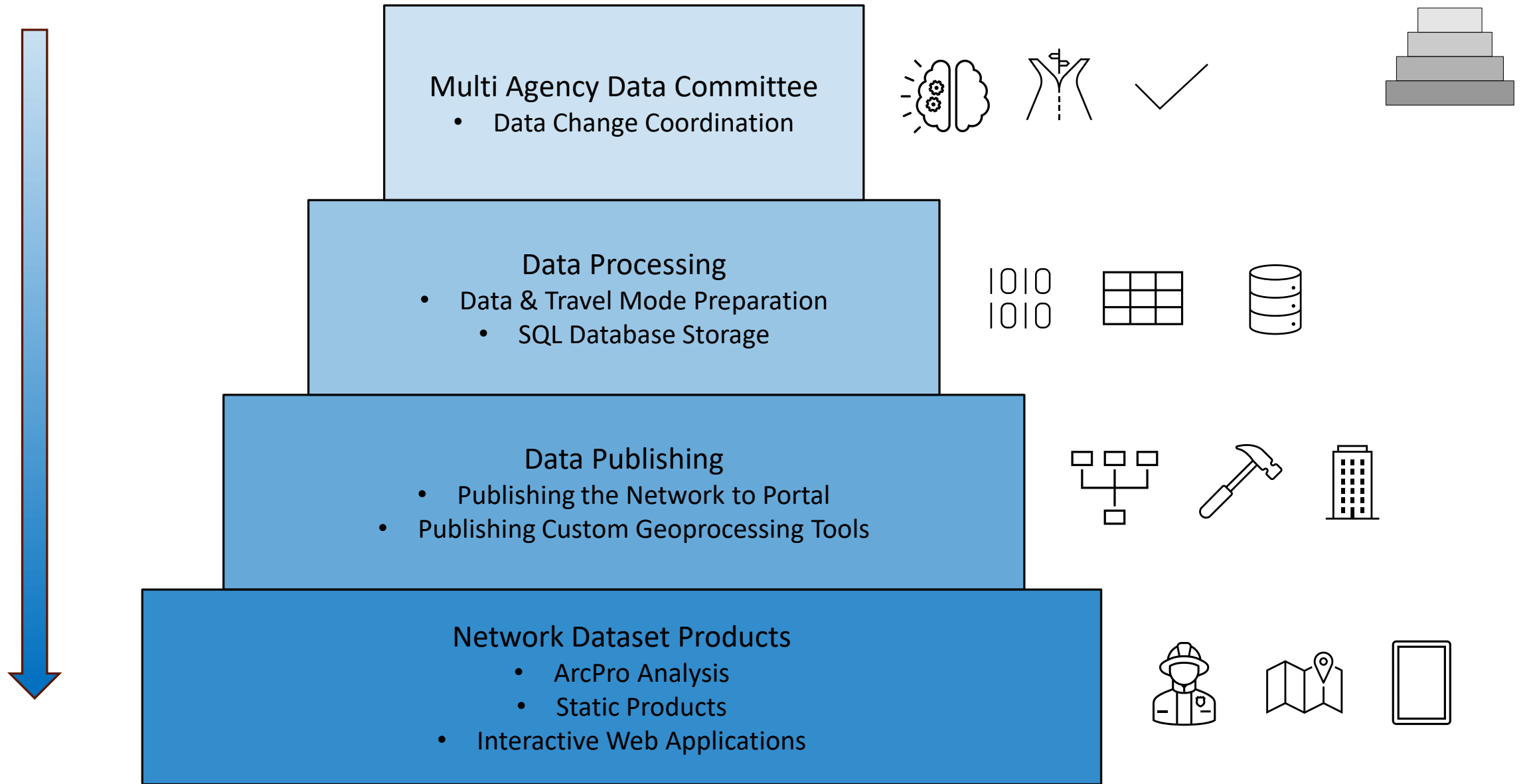
We can manage local conditions, control routing rules, and run analysis without consuming external service credits

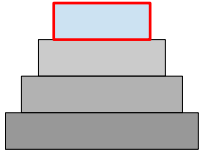
- More Up-to-Date Data
- Dynamic Attribute Control
- Operational Flexibility
- No Credit Usage
- Consistency with CAD



How is this different from Google & Apple?







Multi-Agency Data Committee

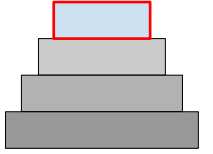
■ Who

- Department of Public Safety Communications
- Department of Information Technology - GIS Data Steward
- Police Department
- Fire & Rescue Department

■ What, How, and When

- Examines roadways & addresses and advises changes
- Meets monthly
- Deadlines determined by lockdown date for CAD Map Roll

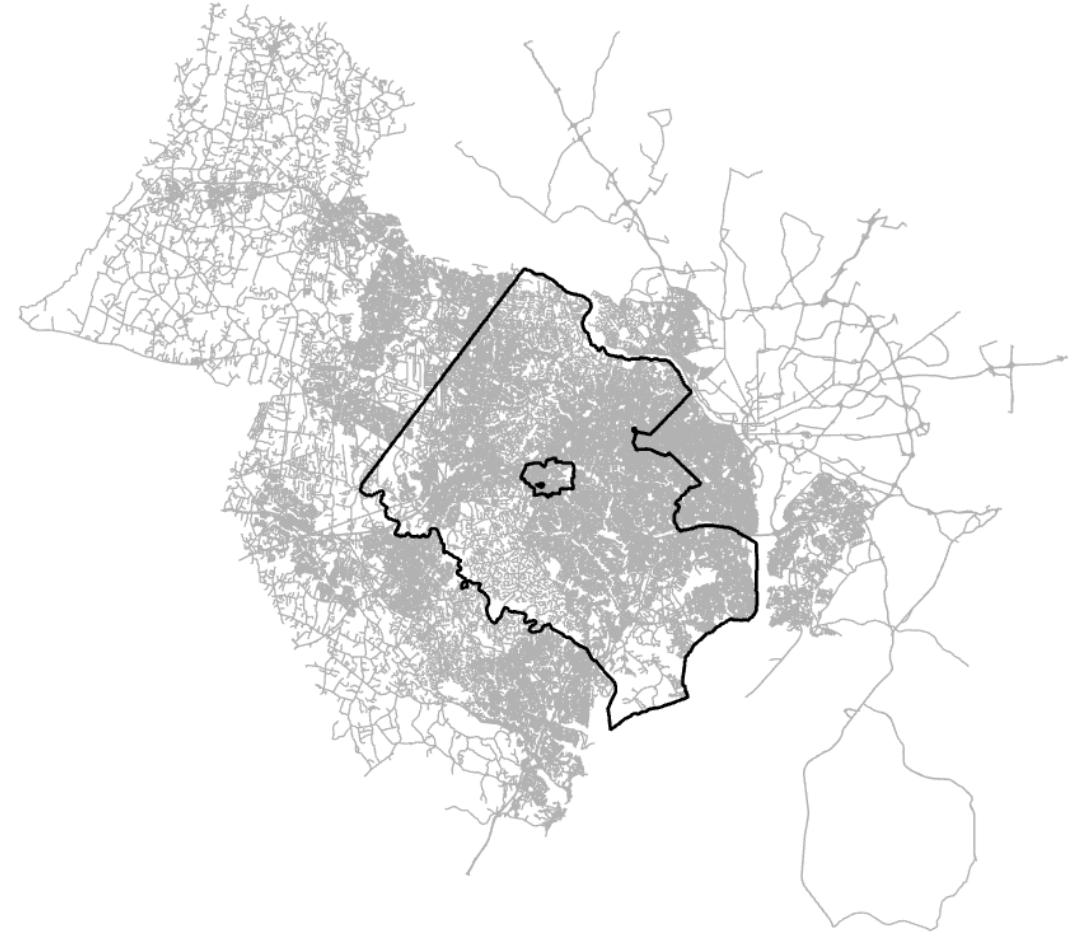


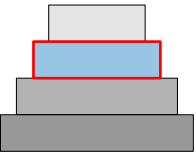


Authoritative Roadway Data

- Roadway Layer View with Comprehensive Schema

- Roadway Name
- Roadway Type
- **Road Class**
- Maintenance Responsibility
- Routing
- **Emergency Use**
- Clearance (height)
- Width
- Weight
- **CAD Speed**
- **One-way**
- **Time Factor**
- **Fire Use**





Data Preparation

Legacy ArcMap workflow

HOW DO I??

Set up Network Analyst for ArcGIS ArcMAP 10.4 - 10.8
By Keg Good, Fairfax County, Fire-IT.

These are the basic instructions for creating a street network. For public safety, the additional steps to eliminate un-routable streets are listed at the end of this document.

- 1) Export TR_Roadway_VW from the EGDB - you must be a FIRE editor (5/2024 this may not be true anymore) (GISMGR.TR_ROADWAY_VW) into a file geodatabase. For Fire use G:\projects\fd\h\q\data\OLDRoadway\... It's not OLD today but with the next update the data will be OLD. Export using ArcCatalog works best.
- 2) Make a dated copy of the original data. (EGDB data will be overwritten after every lockdown) Note: coordinate system is: NAD_1983_StatePlane_Virginia_North_FIPS_4501_Feet.
- 3) Make another copy for your working edits. Sept 2024 142,150 records before edits. May2024 142,134 records before editing. Feb2024 142,071 records. 11/8/23 141,653 records. 7/14/23 141,165 records to start with. 3/17/2023 141,099 records. 10/28/2022 140,666 6/24/22 140,299 records. 3/2/22 138,487 records in ROADWAYS. 10/22/21 139,996 records. 7/6/21 139,962 4/19/21 139,608 8/5/2020 139,637 records. 10/20/17 100,806 records (add PWC) 2/2018 126,687 records (add Loudoun Co.) 6/2018 130,672 records (Updated Arlington and move PGI) 9/2018 130,719 records. 2/2019 130,762 records. After adding routes from Maryland FS- 6/2019 134,782 6/24/19 134,829 records. 2/2020 136,114 records.
- 4) Edit new data as needed. It is important that you interactively review each category unless you know what is in it. With DIT GIS making edits, you may not be aware of all the latest changes.
 - a) Delete all records where FIRE_ROUTE_USE = NO: 358 record 2/2024. 360 records on 10/28/22 (2 of these on DAR/DTR should NOT be deleted.) 370 records on 6/24/22. ~372 records. 365 as of 2/2020. (many of these are Road_Class = Elevated Intersections.)
 - b) Delete one way streets leading into the network if these streets are not otherwise capped. These will be outside Fairfax around the edges of the network. Easiest done by symbolizing on ONEWAY, then manually reviewing around the edges. As of 9/2024 all one way segments should be capped except Page Mill Bl. 2/20 also a few in PGCO.
 - c) No longer applicable but you might review: RW_NAME = 'XXRESTRICTED'. 3/2023 4 in data. FS423 driveway needs to be kept. Effective Fall 2023, beltway imaginary crossovers were deleted.
 - d) Prince William county has over 300 records of ROAD_CLASS = TRL. Many are school driveways. Most of these are harmless in ARL or PWC. Delete any that do not intersect any roads.
 - e) Delete all but one oneway = "B" (about 40 records). Currently there is one record that can be left near Skyline but do a quick check anyway. The record near the landfill doesn't hurt anything. Record to leave: SKYLINE DR SEGID 5021245 TRANS ID 500221276
 - f) Delete any island streets (streets that do not connect to any other streets). This usually only occurs outside of Fairfax County. Prince William has several trails that do not connect. (Similar to item b above.) There may also be a few island segments due to Oneway data in PG.

141,493 records in 9/11/24. 141,776 records in 5/17/24. 141,713 records in 3/5/2024. 140,957 records 11/8/23. 140,340 records as of 7/14/23. 140,740 records left as of 3/17/2023. 140,328 records left as of 10/28/22. 139,927 records left. 6/24/22. 138,445 3/2022 from Roadways. 139,623 records as of Oct 2021. 139,319 records as of June 2021. 139,265 April 2021. 139,455 records as of Dec 2020. 110,429 records as of Oct 2017. 126,313 records Feb 2018. 130,295 records June 2018. 130,343 records 9/2018. 130,388 records 2/2019. 134,415 records 6/2019. 134,468 records 6/25/19 2/21/2020 135,741 records. 8/6/2020 134,274 records including one road cap added.

- a) Add travel time. Add DriveTime column. Double. (Float Precision = 6, scale = 4), alias to Minutes.

G:\projects\fd\h\q\How Do I\Network Analyst\Network Analyst\2020DEC.doc last edit: 2/26/2024 Page 1

ig saved expression or formula below. This is the segment. Depending on how your original data may Id names below may need modification. $[ID] * .0113636 * [TIME_FACTOR]$ eTime.cal (on Keg's OneDrive - good@OneDrive - Fairfax County sAndQueries and in a cannot calculate the records using the alias Minutes. e CAD_SPEED. ROADWAYS uses

DriveTime. If there are, review the segment lengths calculate. There are some very low values such as at the script performed well.

n or not.

To Network Analyst an "N" means no travel is data is FT for restrictions (no travel is allowed

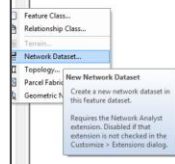
2 way blank N No way delete record, or make blank B

It must be in a data set. Create a new data set, then new data set. Coordinate systems must S_4501_Feet

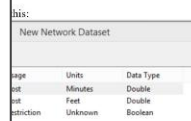
Plane_Virginia_North_FIPS_4501_Feet

py. This step is always a struggle.

m. Customize > Extensions and add the tool bar. If sw Data Network Wizard. Right click on the data set.



0.1. Make sure your input data (streets) are selected. Next. Model Select None for elevation. Next.



name Oneway to ONEWAY, and rename Length to Shape_Length

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NEWAY.

ur intuition.

arent).

u revised the original Oneway attributes.

Next.

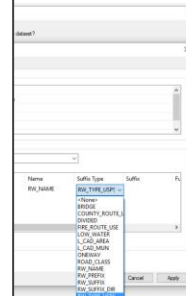
g directions settings for this network dataset?

Directions settings or you can click the Directions button g. You can change the direction settings now, or you a network dataset has been created.

. You don't need

hen hit: Directions. or Name, RW_PREFIX for Prefix, and RW_SUFFIX

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ce Area Index. In order to generate lines and detailed polygons,

is this:

or whatever your name is) % Dataset

whatever your name is.)

Next) (or whatever your name is)

-Prohibited

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tion

whatever your name is)

or whatever your name is)

whatever your name is)

whatever your name is)

Analyst\2020DEC.doc last edit: 2/26/2024 Page 6

whatever your name is)

SPS (or RW_Suffix)

d to build. Hit Yes. Building takes just a couple minutes. You mother file ending with ND_Junctions for a total of 3 files in your

Type File Geodatabase Network Dataset Items File Geodatabase Feature Class File Geodatabase Feature Class

it: drag the NetworkND into your mxd'sTable of

n and add the toolbar. The name of your network will automatically Network2018MarchAllStreets. If you make sure to change to this one for testing.

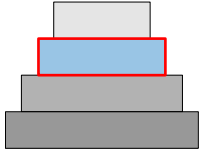
ce sure oneways are being interpreted correctly. The easiest way to XD and do simple routes involving interstates & divided highways.

WAY_VW, use the ROADWAY data and join the using SEGID. Copy result into a new feature class. This is needed use the ROADWAYS data and be prepared to add back in Safety can go but the public cannot. These include selected s leaving fire stations, etc.

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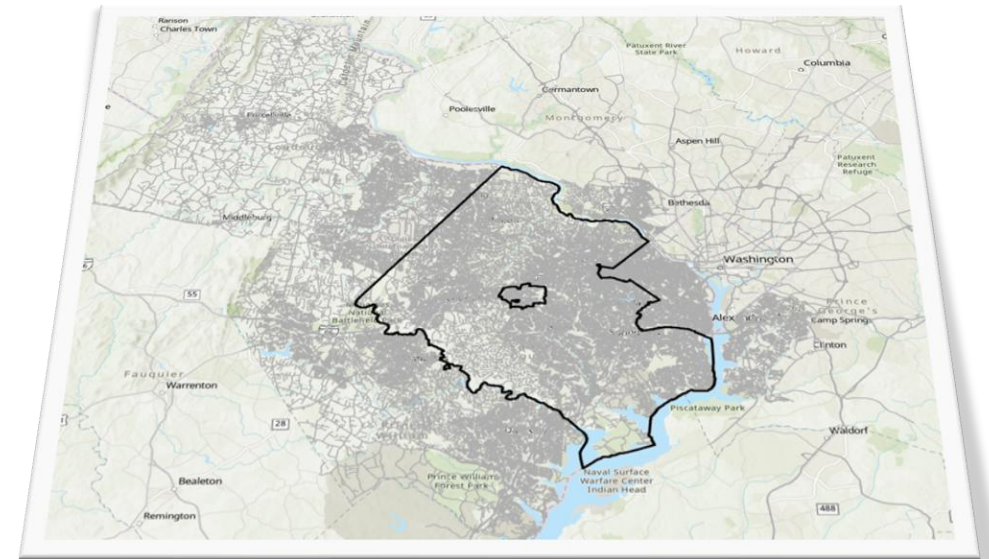
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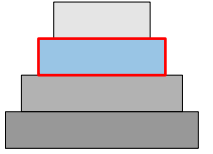


Data Preparation

- The schema defines how the network understands each roadway segment
- Steps:
 - Modify existing fields
 - Add new fields
 - Standardize and configure roadway attributes for fire response
- Goals:
 - Standardize & automate
 - Model builder



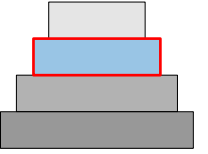
Data Preparation: Existing Fields



- Identify centerlines that are not used for fire apparatus routing
 - FIRE_ROUTE_USE = NO
 - Select the results & remove from the data
- Alleys, trails, federal property, etc



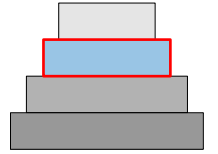
Data Preparation: Existing Fields



- One-way road segments
 - Current set up defines one-ways as yes (Y) or no (N)
 - Change “Y” to “FT” (From To)
 - Change “N” to blank
- Example:
 - Green = origin, Red = destination
 - Complex multi-roadway intersection
 - Route is forced to follow legal direction of travel



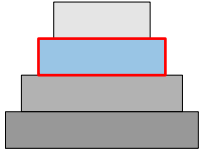
Data Preparation: Existing Fields



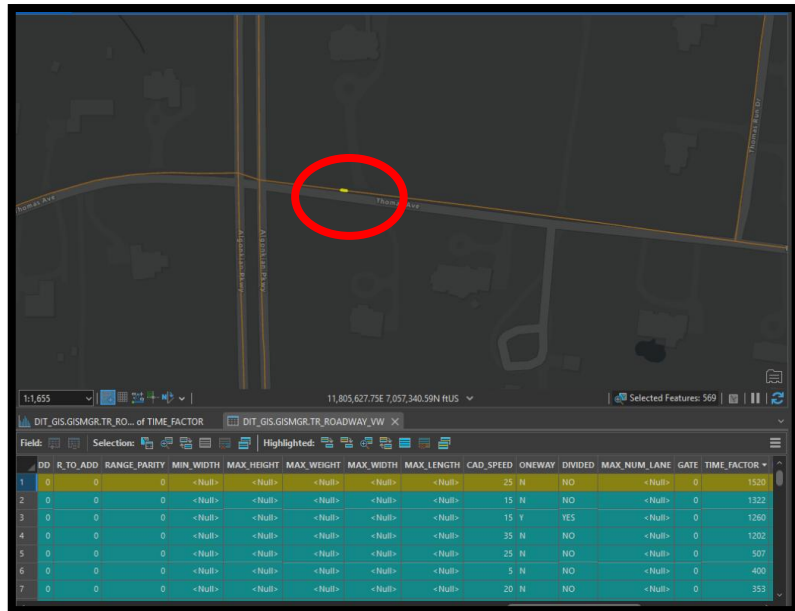
- Future Enhancement: Roadway height, weight & width
 - Identify low clearance and max weight of bridges throughout the county using VDOT bridge and culvert layer (Authoritative source)
 - Update source centerline layer with these values (existing fields MAX_HEIGHT/WEIGHT)
- Width limitation:
 - Cannot currently calculate roadway width with current centerlines & edge of pavement layers
 - Future enhancement
 - Schema is currently set up to hold this data
 - Specs on Vehicles

Field Name	Field Type	Field Length	Field Width	Field Units
VERT_UNDER_LEFT	Text	✓	✓	✓
VERT_UNDER_RIGHT	Text	✓	✓	✓
VERTICAL_CLEARANCE_OVER	Text	✓	✓	✓
VERTICAL_CLEARANCE_UNDER	Text	✓	✓	✓
WIDTH_ROAD_LEFT	Text	✓	✓	✓
WIDTH_ROAD_RIGHT	Text	✓	✓	✓
POSTED_DATE	Date	✓	✓	✓
EV2_GROSS_WEIGHT	Double	✓	✓	Numeric
EV2_SINGLE_AXILE	Double	✓	✓	Numeric
EV3_GROSS_WEIGHT	Double	✓	✓	Numeric
EV3_SINGLE_AXILE	Double	✓	✓	Numeric
EV3_TANDEM_AXILE	Double	✓	✓	Numeric
POST_SIGN_TYPE	Text	✓	✓	✓
POST_CAPAC_EV3	Short	✓	✓	Numeric
POST_CAPAC_EV2	Short	✓	✓	Numeric
POST_CAPAC_EV3	Short	✓	✓	Numeric
CUSTODIAN	Text	✓	✓	✓
OWNER	Text	✓	✓	✓

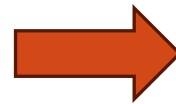
Data Preparation: Roadway Schema Additions



- Add DriveTime field
 - Provides a drive time value for each centerline segment
 - $\text{DriveTime} = ((\text{!SHAPE_Length!} / \text{!CAD_SPEED!}) * 0.0113636) * \text{!TIME_FACTOR!}$



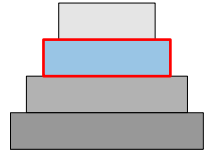
ID	R_TO_ADD	RANGE	PARITY	MIN_WIDTH	MAX_HEIGHT	MAX_WEIGHT	MAX_WIDTH	MAX_LENGTH	CAD_SPEED	ONEWAY	DIVIDED	MAX_NUM_LANE	GATE	TIME_FACTOR
1	0	0	0	<Null>	<Null>	<Null>	<Null>	<Null>	25	N	NO	<Null>	0	1526
2	0	0	0	<Null>	<Null>	<Null>	<Null>	<Null>	15	N	NO	<Null>	0	1322
3	0	0	0	<Null>	<Null>	<Null>	<Null>	<Null>	15	Y	YES	<Null>	0	1260
4	0	0	0	<Null>	<Null>	<Null>	<Null>	<Null>	35	N	NO	<Null>	0	1202
5	0	0	0	<Null>	<Null>	<Null>	<Null>	<Null>	25	N	NO	<Null>	0	507
6	0	0	0	<Null>	<Null>	<Null>	<Null>	<Null>	5	N	NO	<Null>	0	400
7	0	0	0	<Null>	<Null>	<Null>	<Null>	<Null>	20	N	NO	<Null>	0	353



Data Preparation: Roadway Schema Additions



UNDER DEVELOPMENT



$$\text{DriveTime} = ((\text{!SHAPE_Length!} / \text{!CAD_SPEED!}) * 0.0113636) * \text{!TIME_FACTOR!}$$

■ Gates & Barriers

- Accurate Travel Time Prediction
- Optimal Route Selection
- Accurate Service Area Coverage

FRD Gates and Barriers

GATE_NAME

IS_BARRIER
No

DELAY_SECONDS
0

TYPE

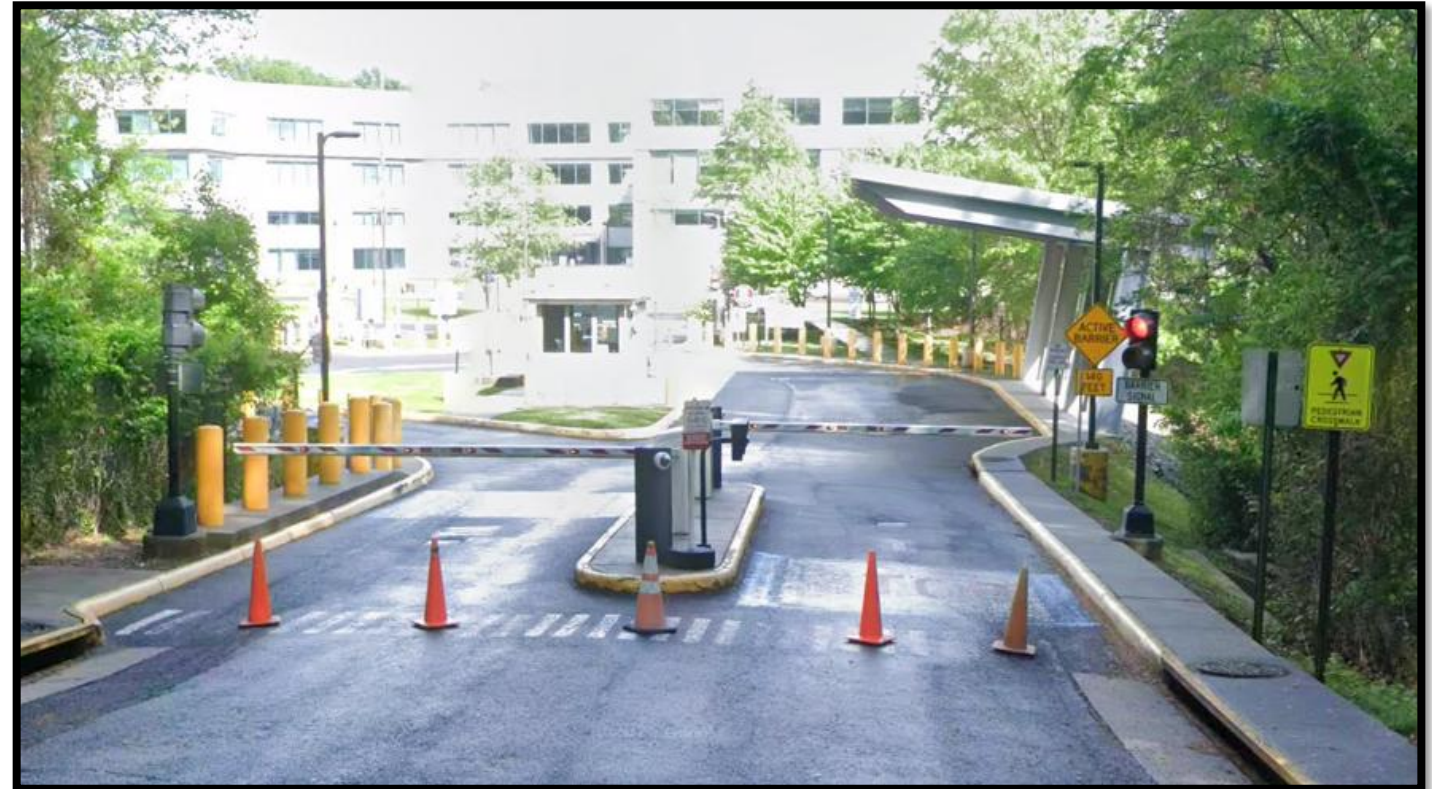
ACCESS_NOTES

ADDITIONAL_NOTES

HOURS

Update Delete

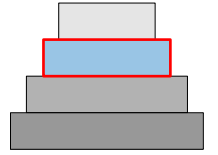
GATE_NAME	IS_BARRIER	DELAY_SECONDS	TYPE	ACCESS_NOTES	ADDITIONAL_NOTES	HOURS
	No	0				



Data Preparation: Roadway Schema Additions

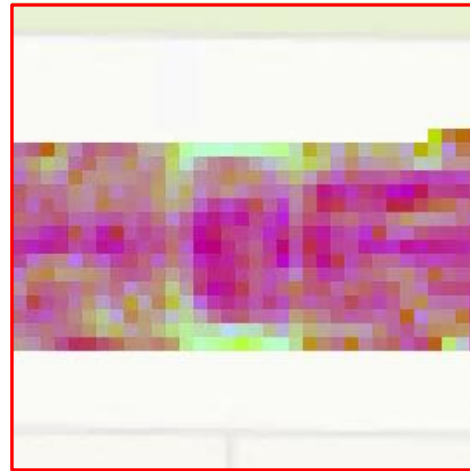


UNDER DEVELOPMENT

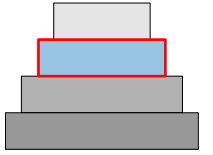


$$\text{DriveTime} = ((\text{!SHAPE_Length!} / \text{!CAD_SPEED!}) * 0.0113636) * \text{!TIME_FACTOR!}$$

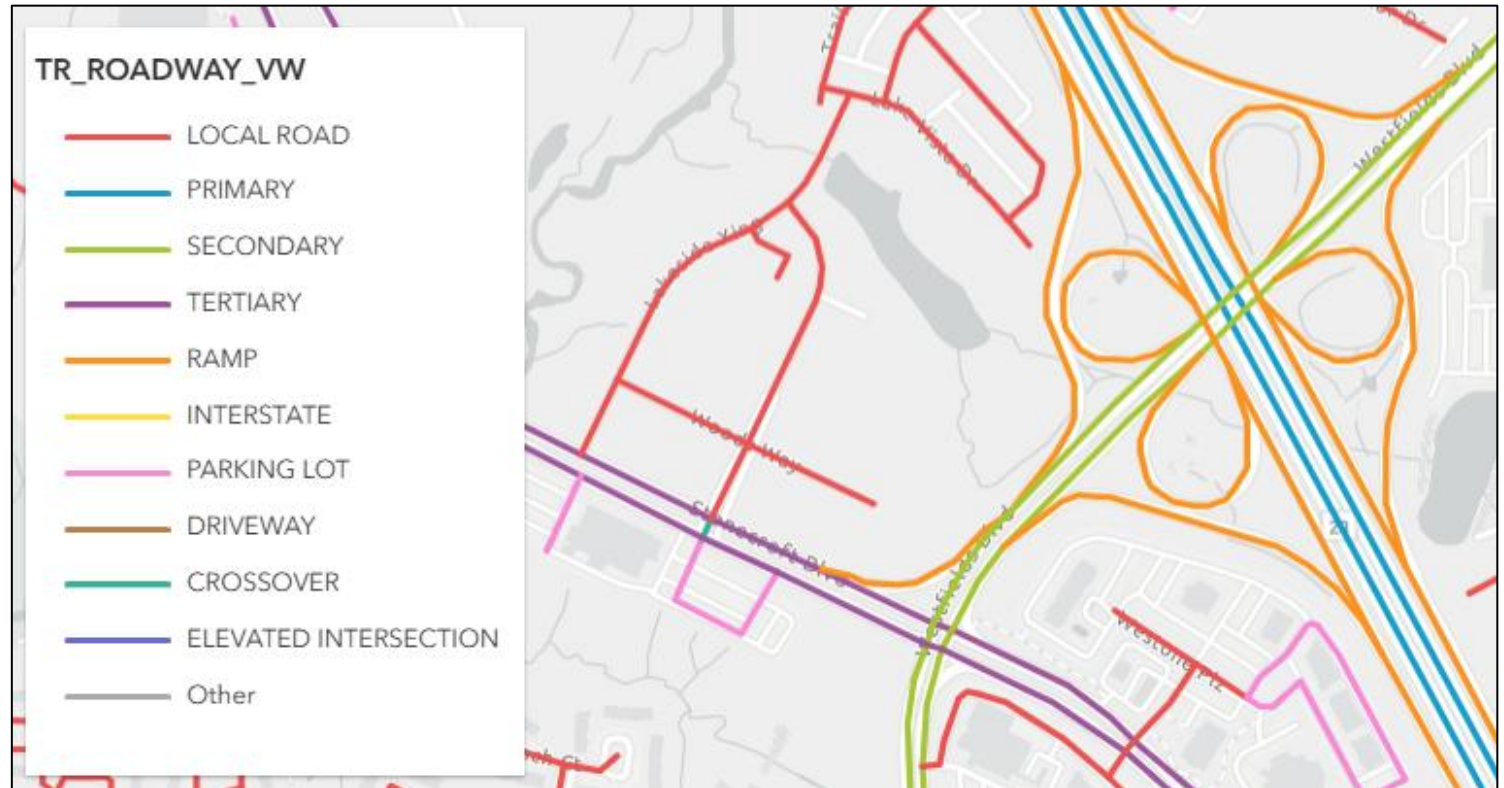
- Speed Bumps & Humps
 - LiDAR Traffic Detection
 - Automated Speed Bump ID
 - Time-Delay Routing Optimization



Data Preparation: Roadway Schema Additions

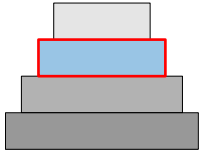


- Add HIERARCHY field
 - Using existing ROAD_CLASS field
 - 1 = INT, HWY, RMP
 - 2 = PRI, SEC, TER
 - 3 = LOC
 - 4 = Other

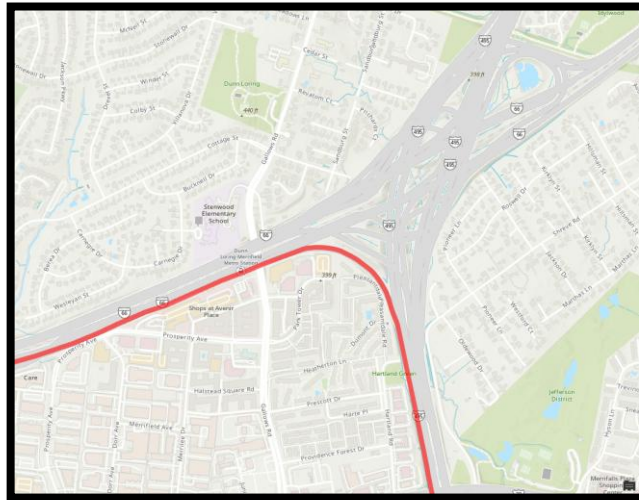


Road Class representing Hierarchy

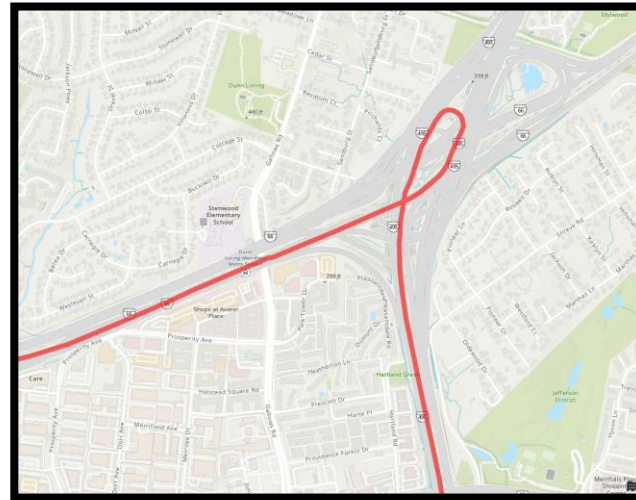
Data Preparation: Roadway Schema Additions



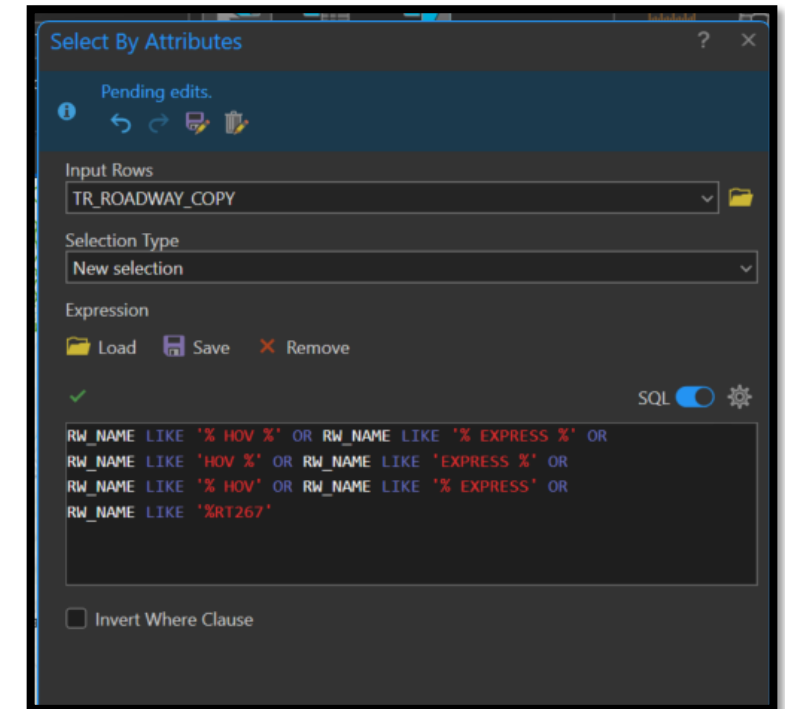
- Add IS_TOLL field
 - Selected all centerlines that are defined as EXPRESS or toll (987 segments)
 - Selected centerlines = 1 is EXPRESS/toll
 - All other centerlines = 0 is not EXPRESS/toll

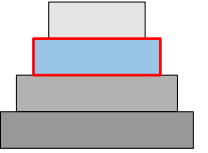


Avoiding EXPRESS/Tolls



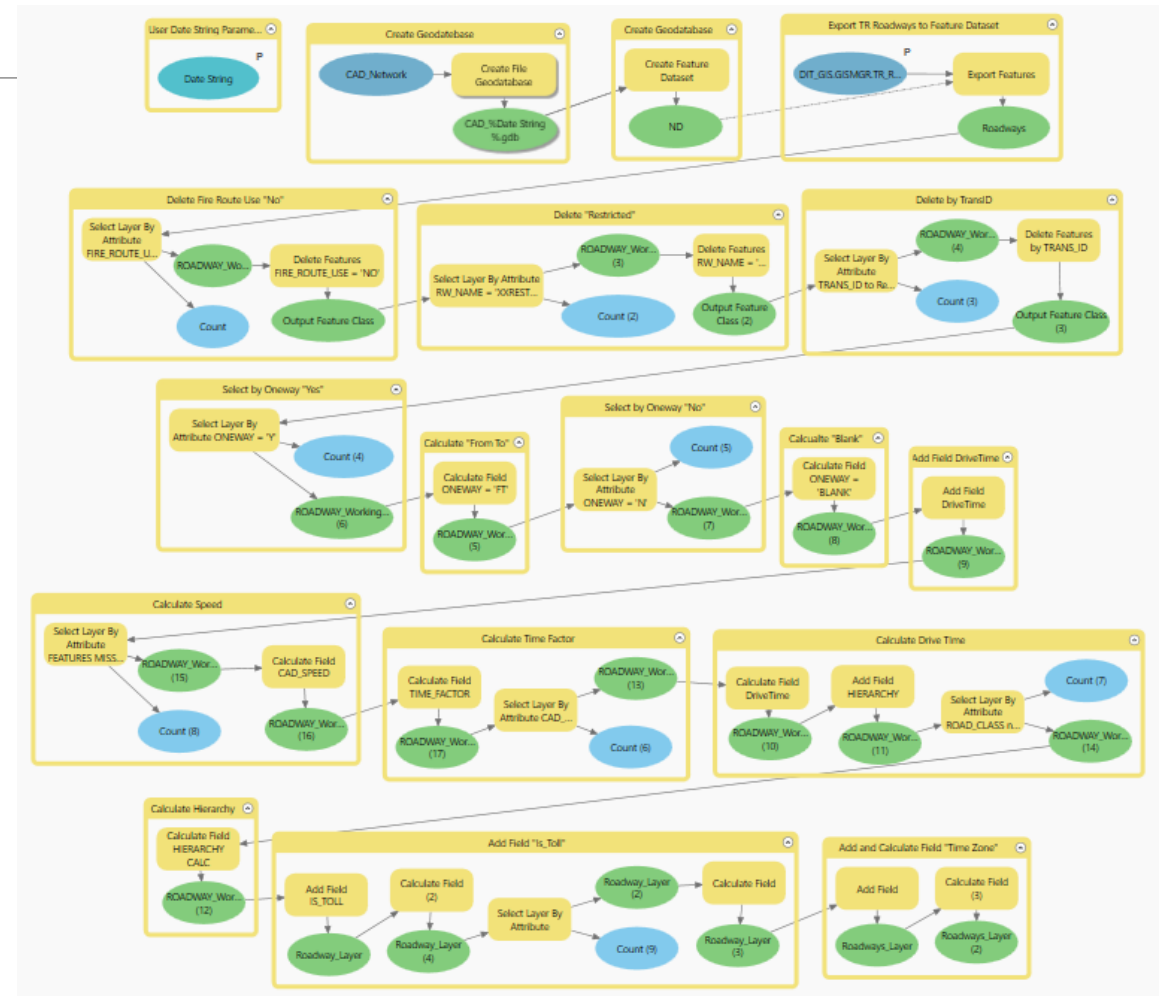
Using EXPRESS/Tolls

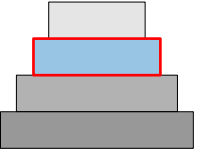




Data Preparation

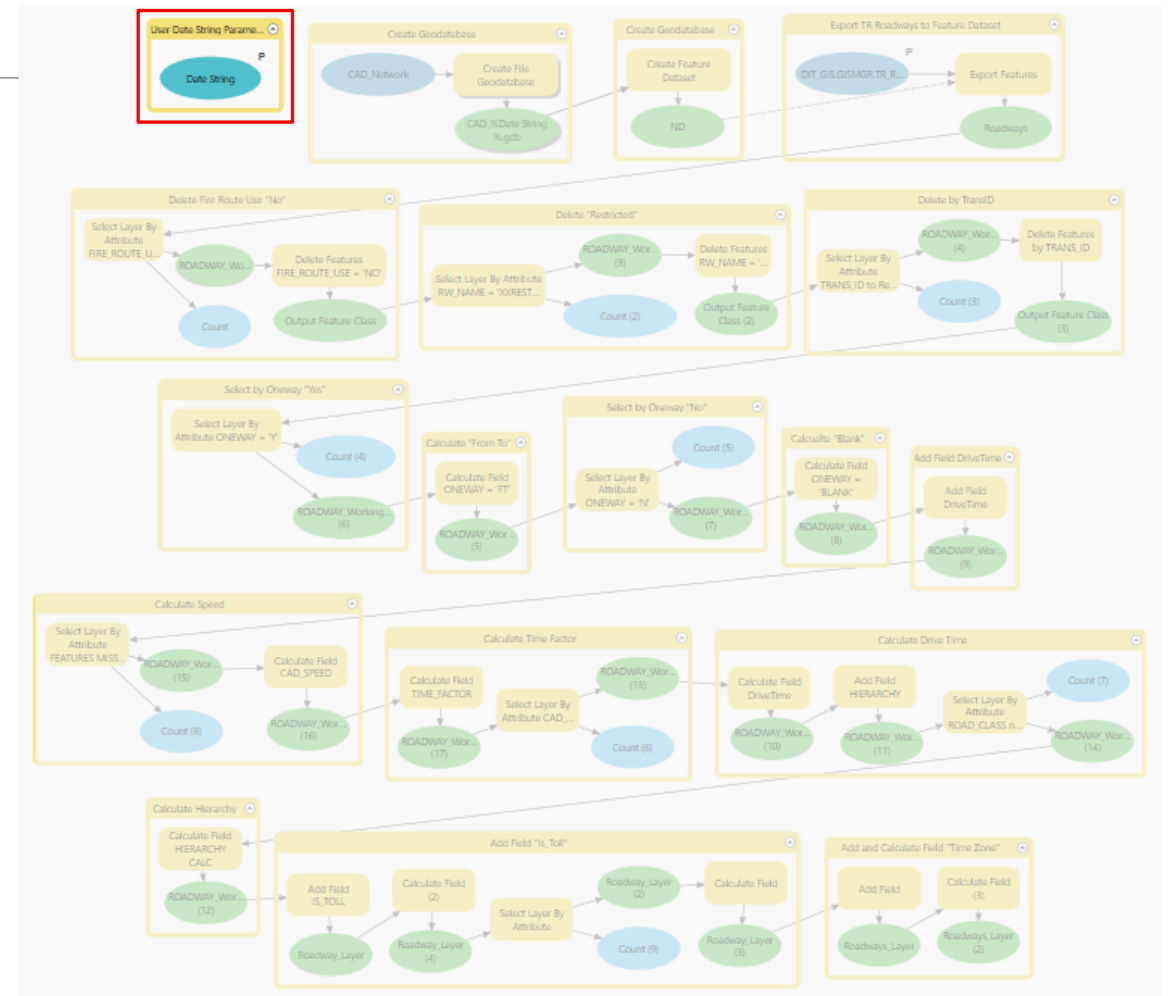
- Using ArcPro Model Builder to streamline and automate data preparation
- Automation reduces staff time
- Automation ensures consistency

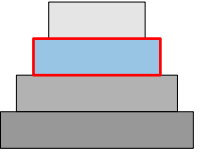




Data Preparation

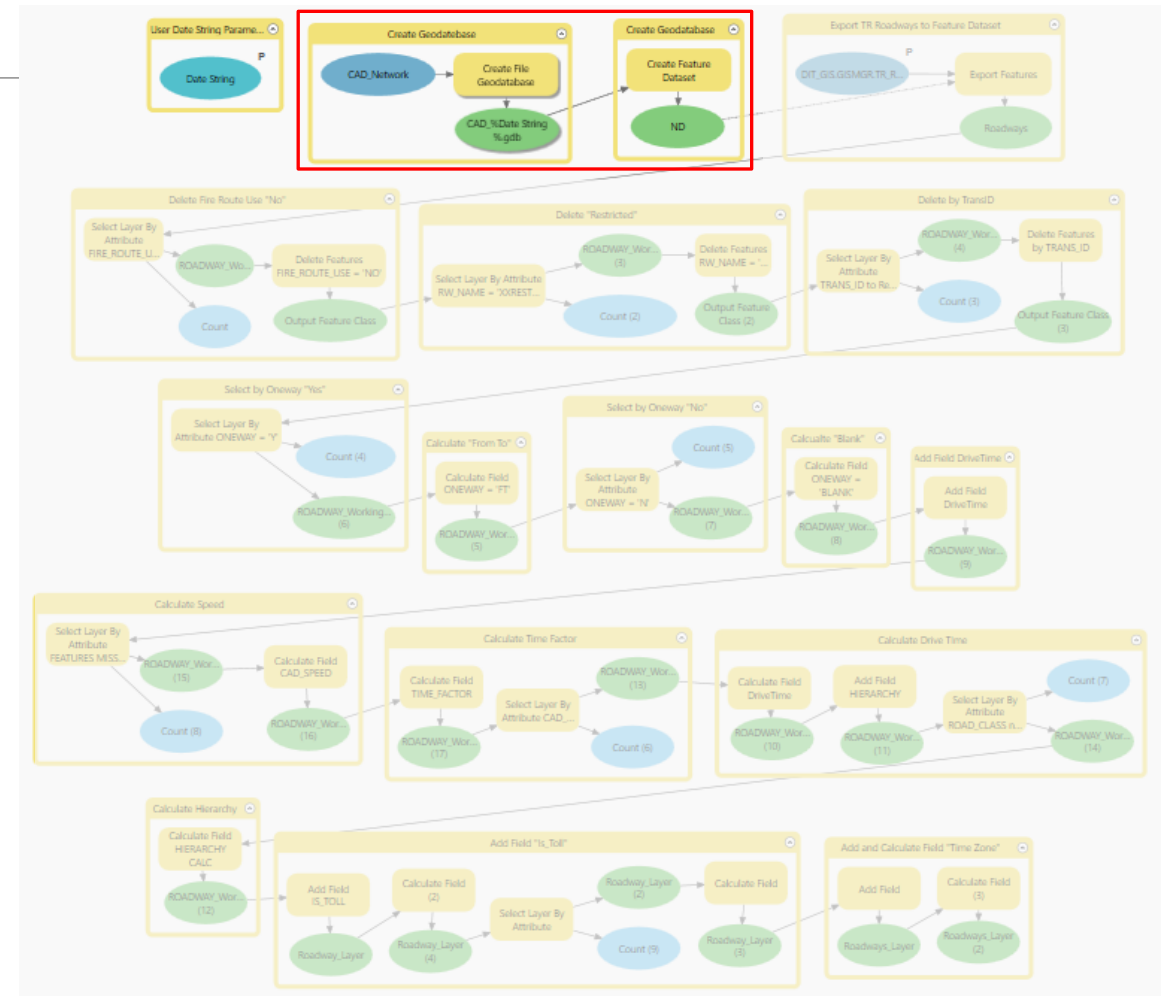
- Parameter - Date
- Create Geodatabase & Feature Dataset
- Copy Roadway Dataset
- Delete Features
 - Restricted Use
 - By Transportation ID
- Calculate
 - One-way
 - Speed
 - Time
 - Hierarchy
 - Toll



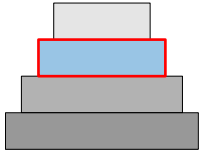


Data Preparation

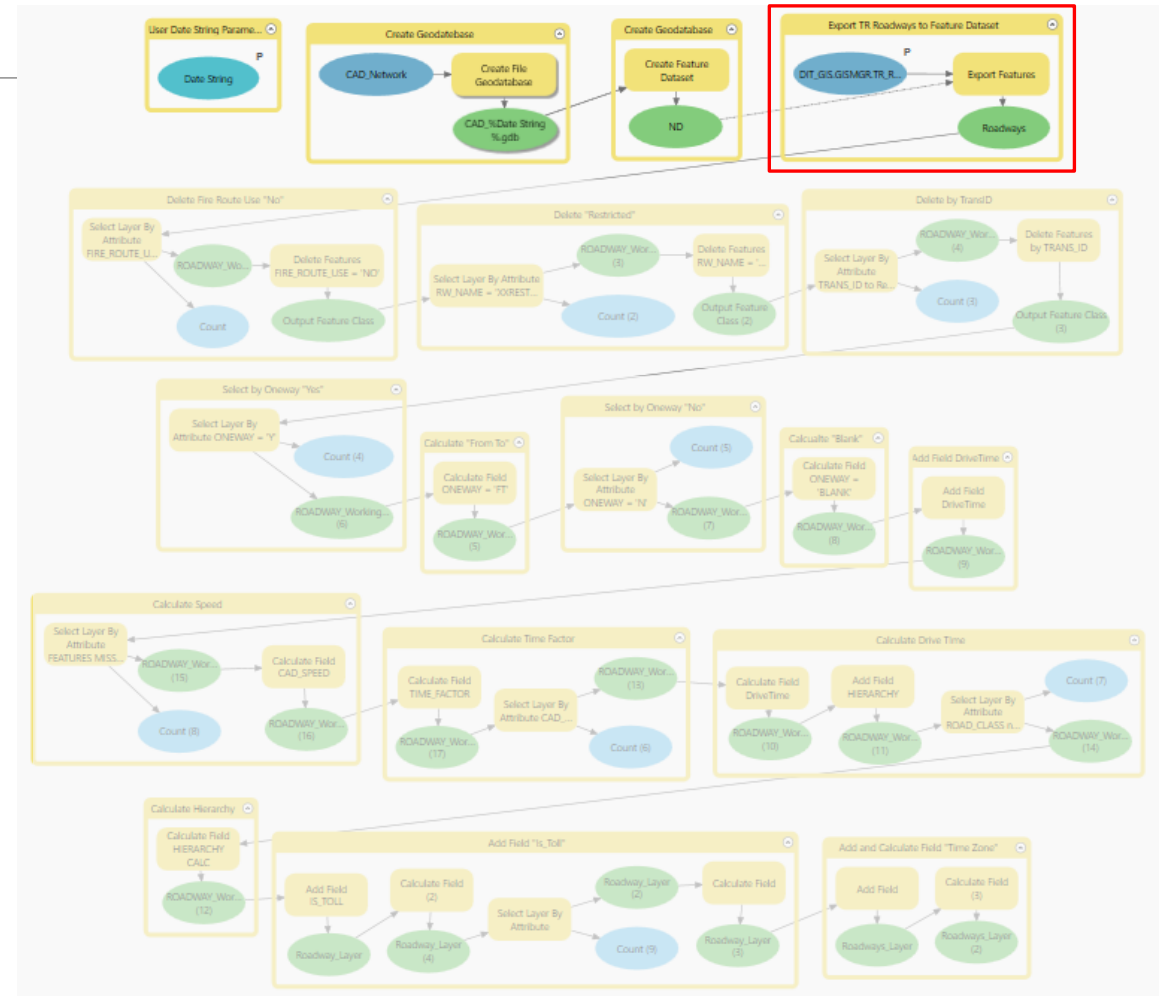
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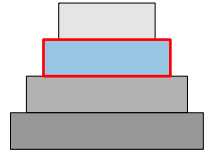
Data Preparation



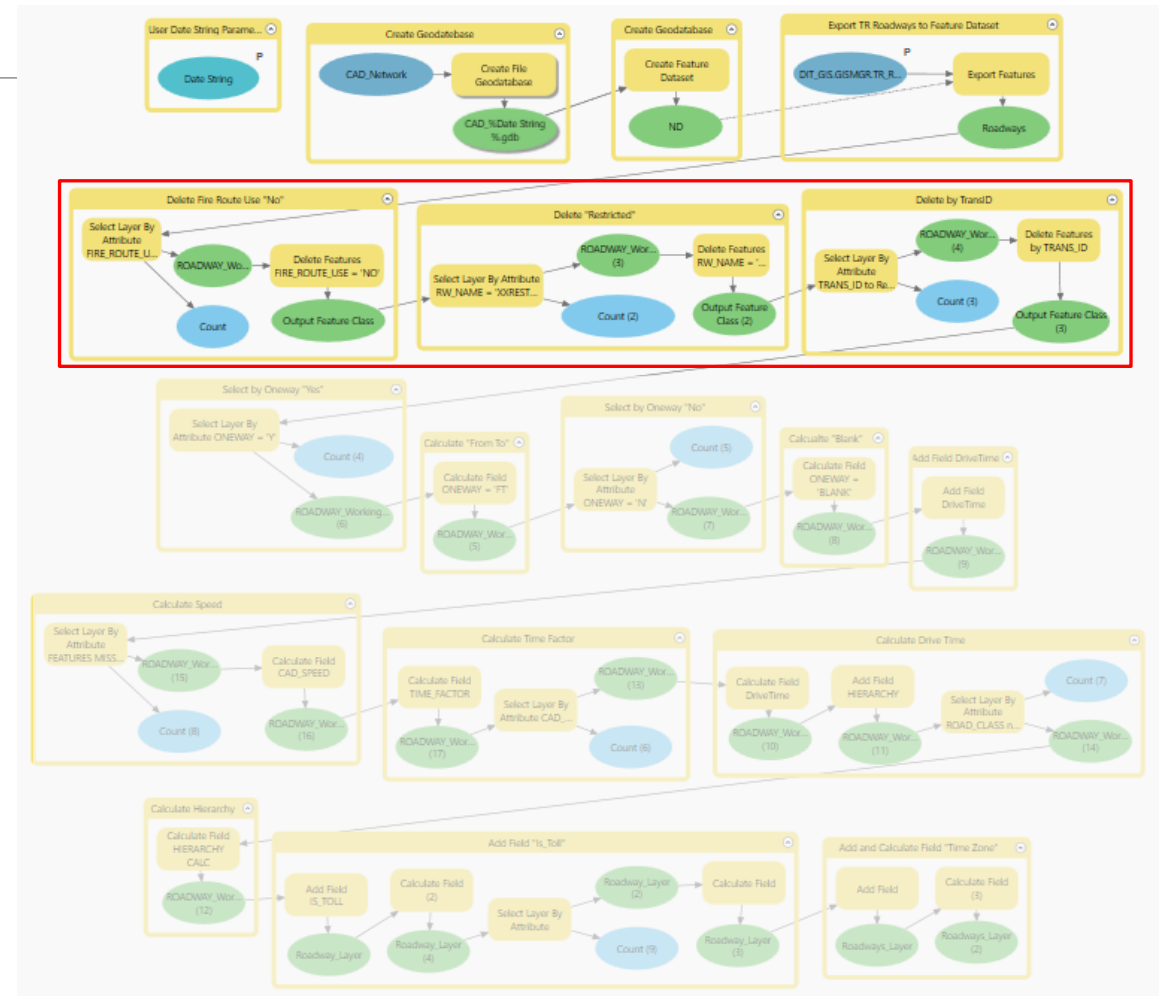
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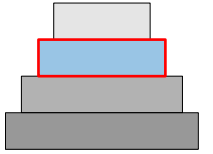
Data Preparation



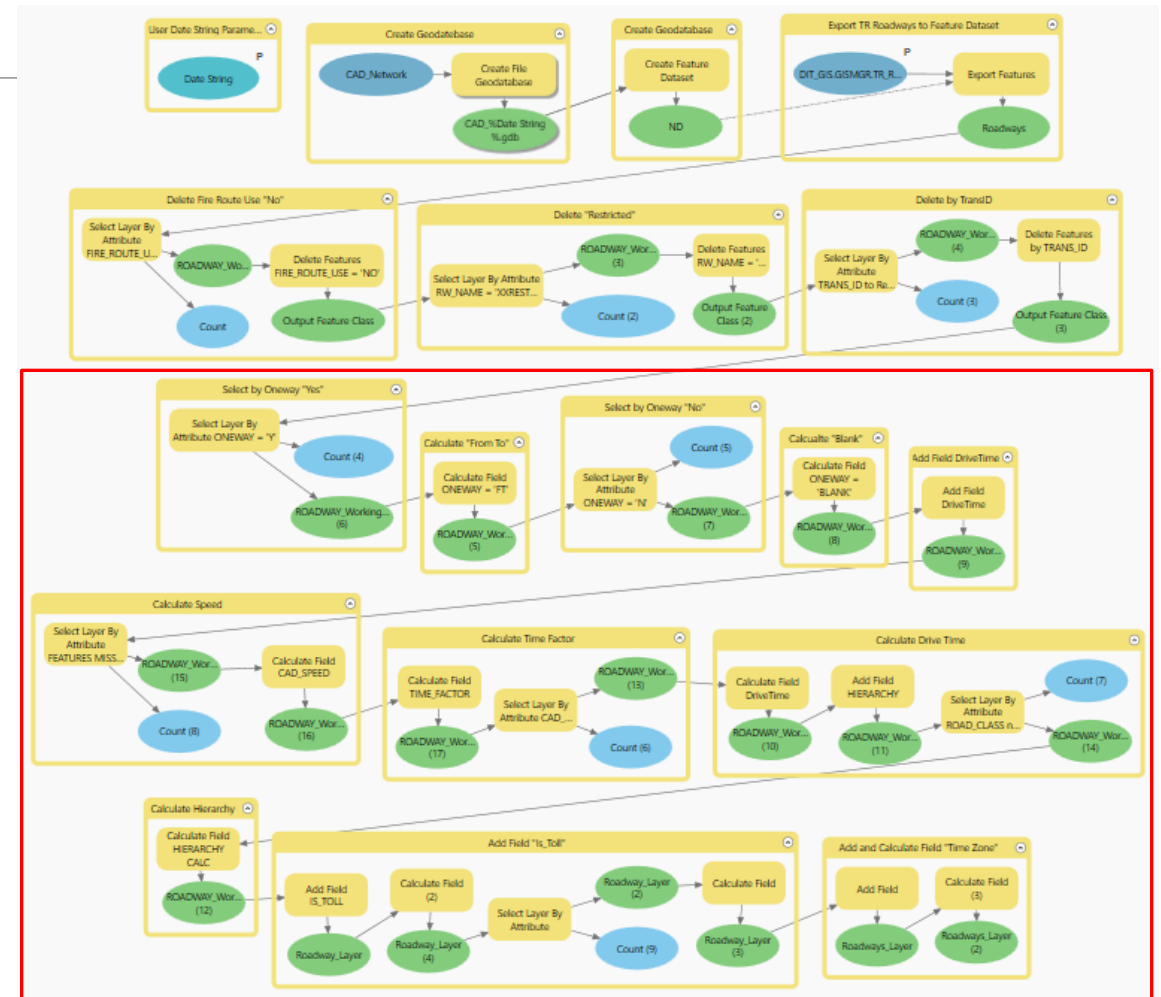
- Parameter - Date
- Create Geodatabase & Feature Dataset
- Copy Roadway Dataset
- Delete Features
 - Restricted Use
 - By Transportation ID
- Calculate
 - One-way
 - Speed
 - Time
 - Hierarchy
 - Toll

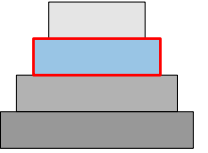


Data Preparation



- Parameter - Date
- Create Geodatabase & Feature Dataset
- Copy Roadway Dataset
- Delete Features
 - Restricted Use
 - By Transportation ID
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 - One-way
 - Speed
 - Time
 - Hierarchy
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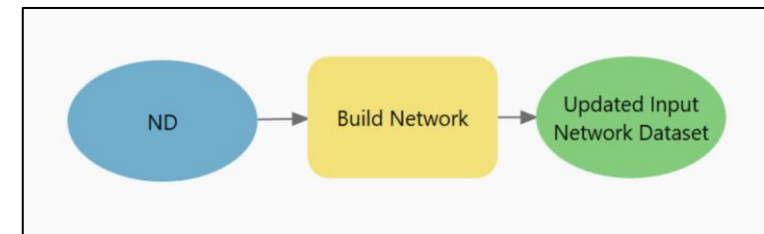
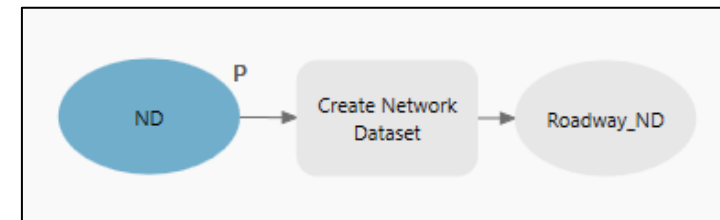
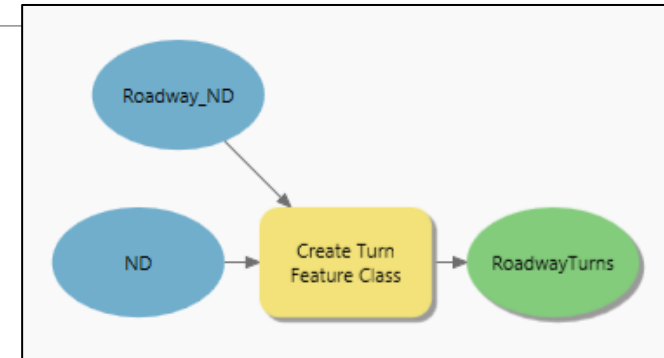




Data Preparation

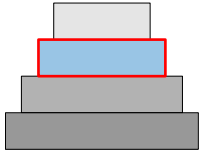
■ Three additional models

1. Create Turns Feature Class
2. Create Network Dataset
3. *Build Network Dataset**

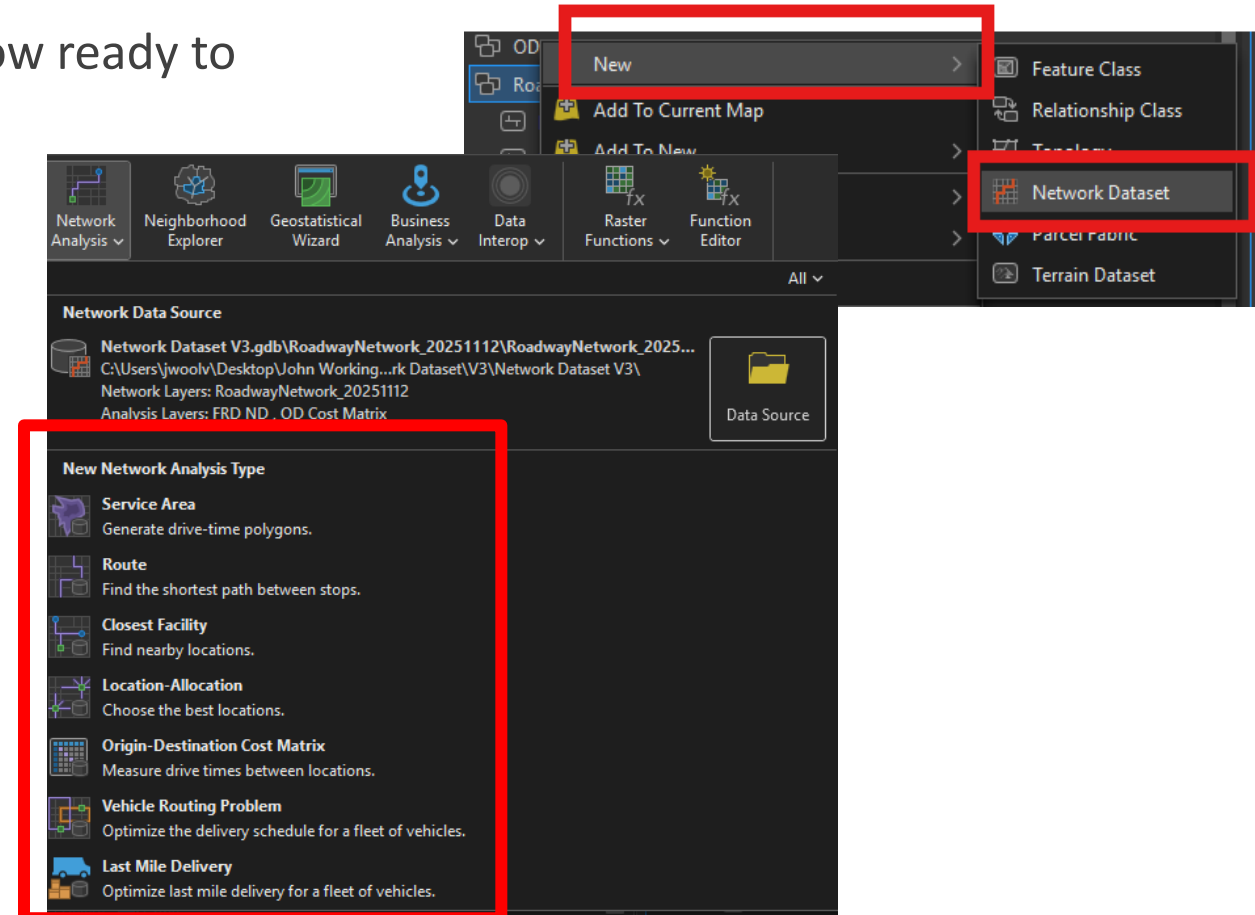


**Building the network is a quick tool that may not need a model*

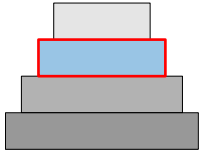
Data Preparation: Build the Network Dataset



- With schema changes complete, the data is now ready to be turned into a network dataset
- Operationalizes the schema design
 - Translates attribute fields into route behavior
 - Configure travel modes, restrictions, & costs
 - Enables network tools, such as:
 - Service Area
 - Route
 - Closest Facility
 - Origin-Destination Cost Matrix



Network Dataset Configuration: Travel Modes



- Predefined configurations within the dataset that define how travel occurs

- Controlled by:
 - Restrictions (one-way streets, height/weight limits, EXPRESS/tolls)
 - Costs (travel time, roadway hierarchy, turns)
 - Impedances (time, distance)

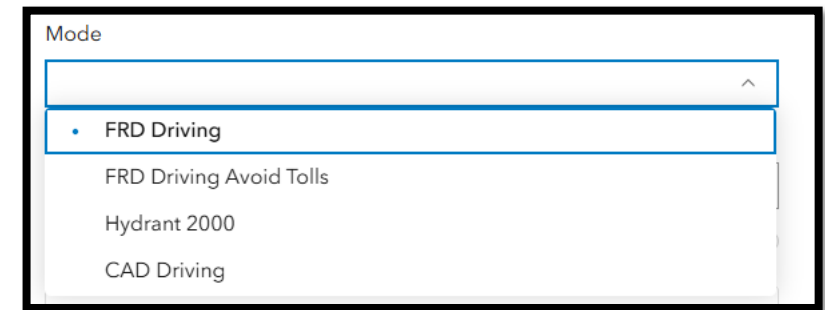
- Importance

- Consistency – uses the same routing rules
- Realism – follow "real-world" restrictions
- Flexibility – switch between modes
- Informed Decisions – compare different routing scenarios

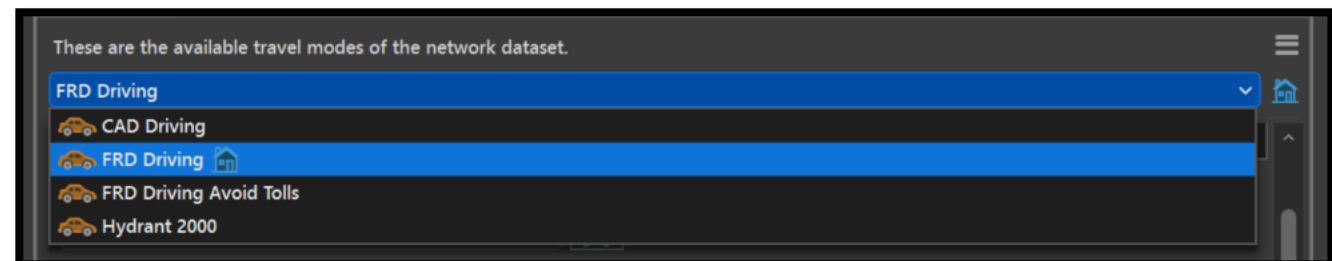
- Travel Modes Types:

- FRD Driving
- Avoid Tolls
- Hydrant 2000 (no one-way restrictions)
- CAD Driving (no turn costs or hierarchy)

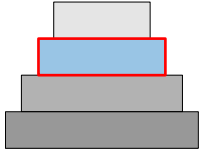
Portal apps



ArcGIS Pro



Network Dataset Configuration: Costs



- Goal: Reflect “realistic” emergency maneuver times

- Turns (seconds)

- Lower than normal driving, but not a zero delay (not realistic)
 - Left turns > Right turns
 - Reverse = heavily penalized

- Hierarchy

- Plays big role in turn costs
 - ex: Intersections at major roads = higher time cost
 - More realistic emergency routing
 - Higher speeds and fewer interruptions = efficiency
 - Avoid local road conflicts until necessary (more stops, tight turns)
 - Keeps routes logical and operationally realistic

Turn Category: DriveTime [Default]

These are the properties of this turn category evaluator.

Turn Azimuth Range

Configure the azimuth angle range that defines both 'Right' and 'Left' turns. A turn rotating less than the 'Min Azimuth' is classified as 'Straight Ahead'. A turn rotating more than the 'Max Azimuth' is classified as 'Reverse'.

Min Azimuth: 30
Max Azimuth: 150

Road Classes

Use Default Ranges from Hierarchy (selected)

Primary Roads: 1 - 1
Secondary Roads: 2 - 2
Local Roads: ≥ 3

Use Custom Ranges

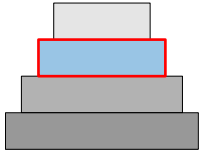
Primary Roads: 1 - 2
Secondary Roads: 3 - 4
Local Roads: ≥ 5

Turn Category Travel Times

From Road Class	To Road Class	Across Road Class	Seconds
Right Turn			
Local	Local	Any	1
Local	Secondary	Any	2
Secondary	Local	Any	3
Secondary	Secondary	Any	2
Left Turn			
Local	Local	Any	4
Local	Secondary	Any	6
Secondary	Local	Any	5
Secondary	Secondary	Any	8
Straight Ahead			
Local	Local	Local	1
Local	Local	Major	8
Local	Local	None	0
Local	Secondary	Any	5
Secondary	Local	Any	3
Secondary	Secondary	Local	3
Secondary	Secondary	Major	9
Secondary	Secondary	None	0

From	To	Across	Seconds
Local	Local	Local	1
Local	Local	Major	8

Network Dataset Configuration: Restrictions



■ Restrictions (Internal vs External)

■ One-way

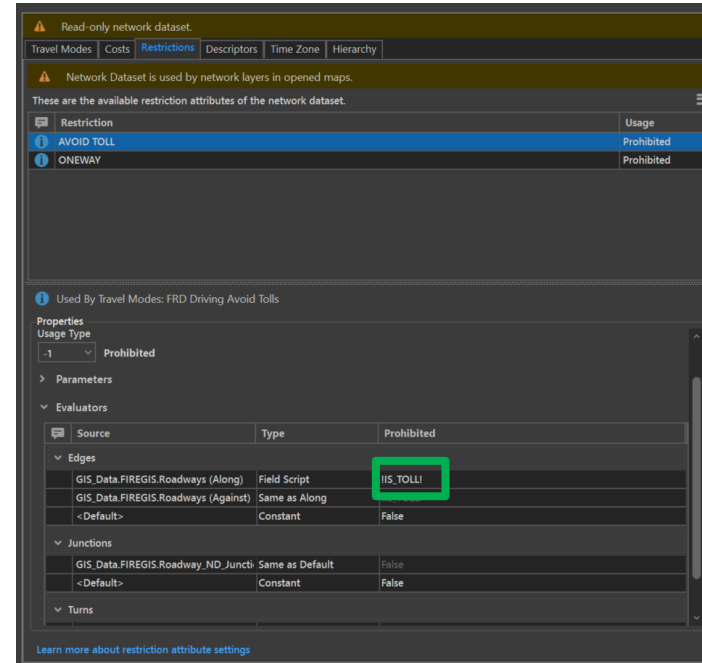
- Recognizes one-way directions
- Routing honors correct side of roadway
- Field script

■ Tolls/HOV/Express

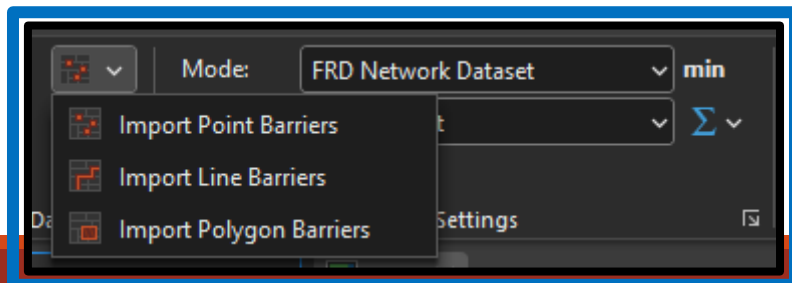
- Proof of concept – routing avoids using tolls/HOV/express lanes
- Field script honors binary field

■ Gates & Barriers

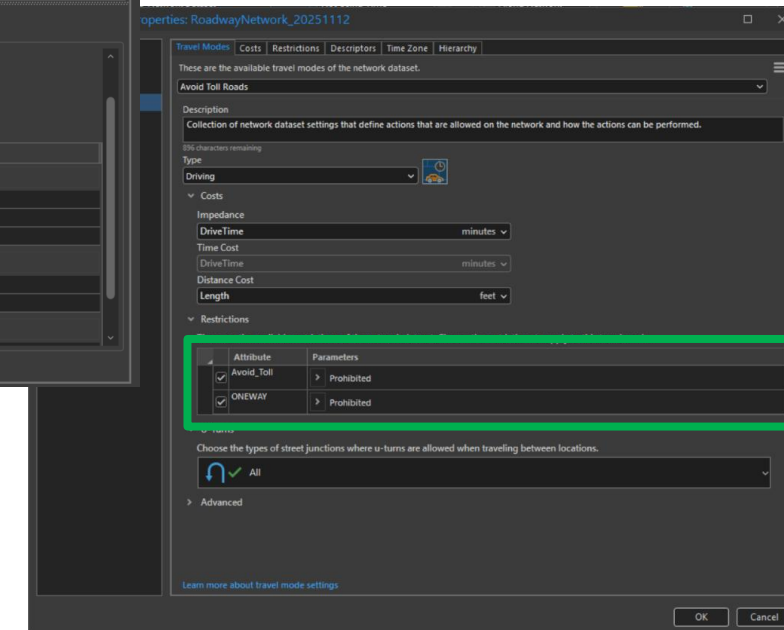
- External layers can be added as restrictions in the analysis ribbon

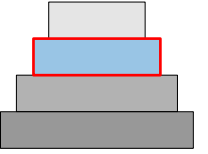


Configure restrictions in the restrictions tab



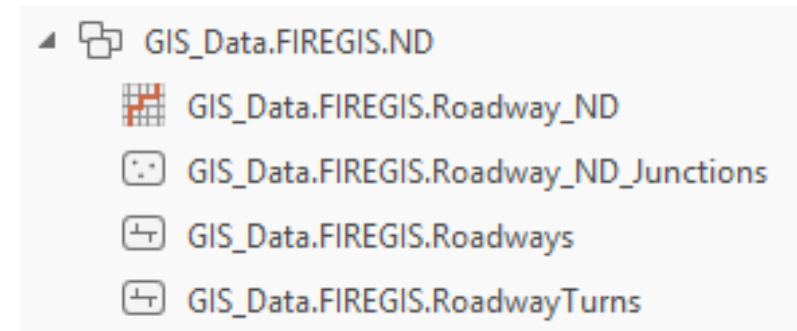
Toggle restriction on/off in the travel modes tab





Network Dataset Storage

- Copy the feature dataset to your SQL database
 - Use admin database credentials
- Includes Travel Modes
- Pre-Built and ready to use
- Users accessing your network should use database reader credentials



A diagram of a four-story building. The building is composed of four stacked rectangular sections. From bottom to top, the sections are: a dark gray base, a medium gray second floor, a light gray third floor, and a white top section. A red rectangle is drawn on the second floor, centered horizontally and vertically within that floor's section.

- OneNote (Microsoft)
- Internal Documentation
- SharePoint (Microsoft)

Avoid Toll/HOV/Express Lane Configuration

Friday, January 09, 2026 4:24 PM

G:\projects\hvd\data\NetworkData\CAD_Network\CAD_20251223.gdb. Below are the step-by-step instructions for adding this restriction and travel mode once the IS_TOLL field is created in the roadways layer.

- Create "IS_TOLL" field in centerline data
- to be able to choose to use the express lanes or not
 - o Field calculate
 - != express/hov
 - Or no toll
 - o This can be configured as a cost in the ND build
 - Query tool [how roads](#)
 - o
 - o

- o RW_NAME LIKE '% HOV %' OR RW_NAME LIKE '% EXPRESS %' OR
RW_NAME LIKE '% HOV %' OR RW_NAME LIKE '% EXPRESS %' OR
RW_NAME LIKE '% HOV %' OR RW_NAME LIKE '% EXPRESS %' OR
RW_NAME LIKE '% TOLL %' OR
RW_NAME LIKE '% NRT287'
- o Selection Results = 994 (as of 12/2023)

- Create "Avoid_Toll" restriction
 - o After rebuilding the ND with the new IS_TOLL field, open ND Properties
 - o Travel Attributes → Restrictions tab → 3 bar menu (red arrow below) → New
 - o Name: Avoid_Avoid_Toll
 - o Usage type = 1
 - o Evaluators:
 - o Edges:
 - Along:
 - Type = "Field script" = IS_TOLL
 - Against:
 - Type = "Same as Along" = IS_TOLL (autofills)
 - o
 - o

Add page

Build Redundant Network Dataset (ArcGIS Pro 3.3)

Data and Documentation:

[\[Project\] > \[Data\] > \[Network\] > \[CAD_Network\] > \[Network\] > \[Network Dataset 2023\] \[11\]](#)

Model for Part 1: CAD Roadway Input

- [G:\project\info\input\data\NetworkDataset\CAD_Network\Network.asxd](#) - Network asxd Roadway Input 2023

Model for Part 2: CAD Create Network

- [G:\project\info\input\data\NetworkDataset\CAD_Network\CAD_Network.asxd](#) - CAD Create Network

Model for Part 3: CAD Create Network Turns

- [G:\project\info\input\data\NetworkDataset\CAD_Network\CAD_Network.asxd](#) - CAD Create Turns

Part 4: Network Configuration

Purpose

This SOP documents the standardized process for preparing CAD roadway network data for fire and rescue operations. The workflow creates a data-restricted geodatabase, exports authoritative roadway data from SQL, removes non-movable and restricted features, and calculates attributes required for routing and response systems.

Scope

This procedure is used to generate a **CAD-ready roadway feature class** for use in routing, travel time analysis, and emergency response systems.

Part 1 - CAD Roadway Data Prep

1. Create Geodatabase "CAD_YYYYMMDD.gdb"
2. Create Feature Dataset "RD"
3. Export roadway copy "Roadway"
4. Export RD_Roadway_V10 from the RDG to a file geodatabase feature dataset.
 - Use Catalog [pane](#) - Right-click > Export Features.

Save in your Fire Projects path (e.g., [G:\project\info\input\data\roadway\](#)).

Build Road Data

1. Clean up enableable segments

- Open an **edit session** in Pro.
- Use **Select by Attributes** or **manual selection** to select:
 - **FFR_ROUTE_SUE = NO**
 - 300 as of 11/5/2025
 - One-way segments at the edges if not capped.
 - Trunk (ROAD_3DASZ = TR) as they do not intersect other roads.
 - 40 as of 11/5/2025, did not select any of them.
 - Some of these seem to not select any prior to Yearmap spring 2025 collection
 - Go through list and confirm/remove fire use field for Cad group?
- Any segments with **RW_NAME = X** (SELECTED if still present) – except verified exceptions.
 - 2 solid (after fire route = no query) as of 11/5/2025
 - kept discovery station 23
 - Deleted the other
 - Remove **Driveway Line** ([Data Explorer](#) view - Right Click Road)
 - Query by the **Driveway** name, find and delete only two segments in **Part1a** is **CAD_MUN** field.

- "Isolated" streets not connected to the network.

• Double-check **enable = 0** segments = keep only valid ones

- Change 1 to **FF**
- Change 10 to **[Isolated]**

Web Maps

Order Used Maps

Web Maps

EMS-GIS Location

The GIS Maps page, maintained by the GIS Branch within the Data Analytics and Strategy Management team, serves as a central hub for accessing a variety of fire and Rescue mapping resources.

Use this page to explore and download PDF maps or interact with dynamic web mapping products designed to support operational planning, situational awareness, and decision-making.

Whether you need static maps for quick reference or web-based tools for deeper analysis, this page provides geographic insights tailored to your needs. Select the location below to submit a map or data request to the GIS Branch.

[Submit Map or Data Request Here!](#)

Wall Maps

Fairfax County Fire & Rescue: 2025 Wall Map

The Fire & Rescue Wall Map is the most comprehensive map available, providing detailed coverage of stations, fire trucks, fire truck areas, battalions, and divisions. It serves as a vital tool for situational awareness, resource allocation, and strategic planning, with an updated version scheduled for release in 2025 to reflect the latest changes.

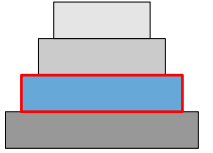
Fairfax County Fire & Rescue: EMS Administrative Coverage

The Fire & Rescue Wall Map is the most comprehensive map available, providing detailed coverage of stations, fire trucks, fire truck areas, battalions, and divisions. It serves as a vital tool for situational awareness, resource allocation, and strategic planning, with an updated version scheduled for release in 2025 to reflect the latest changes.

Fairfax County Fire & Rescue: EMS Supervisor Coverage

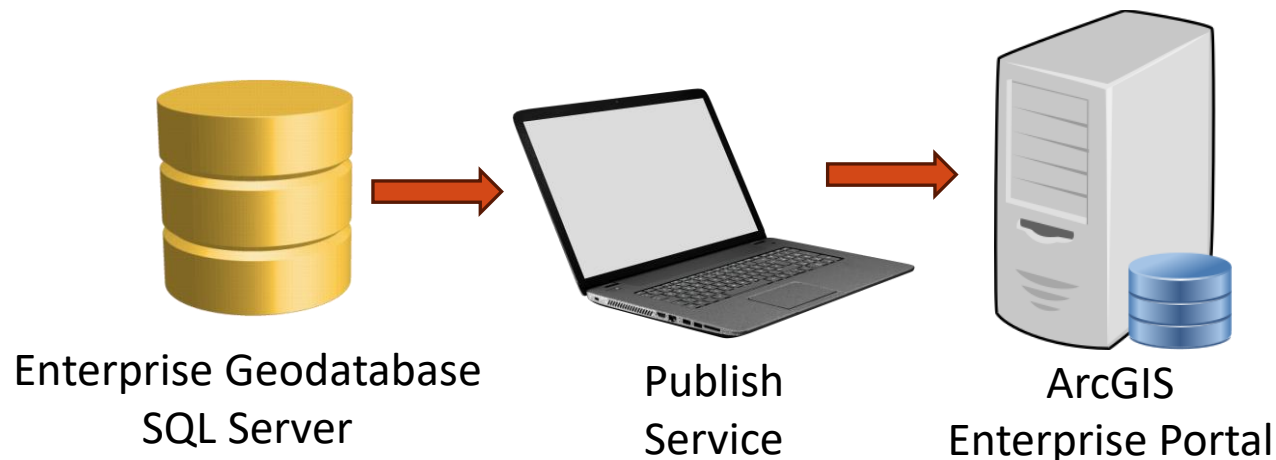
This map provides a straightforward depiction of fire stations and supervisor districts, designed for quick reference. A summary table accompanies the map, listing stations by supervisor district for support coordination.

Publishing the Network Dataset



■ Requirements

- Portal is federated with your server
- Network Analyst license for Server & Portal



Configure routing services

Routing utilities (Synchronous)

Example: <https://webadaptorhost.domain.com/webadaptorname/rest/services/folder/serviceName/GPServer>

Network Analysis Services

Closest facility

Example: <https://webadaptorhost.domain.com/webadaptorname/rest/services/folder/serviceName/NAserver/layerName>

Origin destination cost matrix

Example: <https://webadaptorhost.domain.com/webadaptorname/rest/services/folder/serviceName/NAserver/layerName>

Route

Example: <https://webadaptorhost.domain.com/webadaptorname/rest/services/folder/serviceName/NAserver/layerName>

Service area

Example: <https://webadaptorhost.domain.com/webadaptorname/rest/services/folder/serviceName/NAserver/layerName>

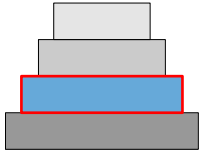
Map Image Layer

Traffic

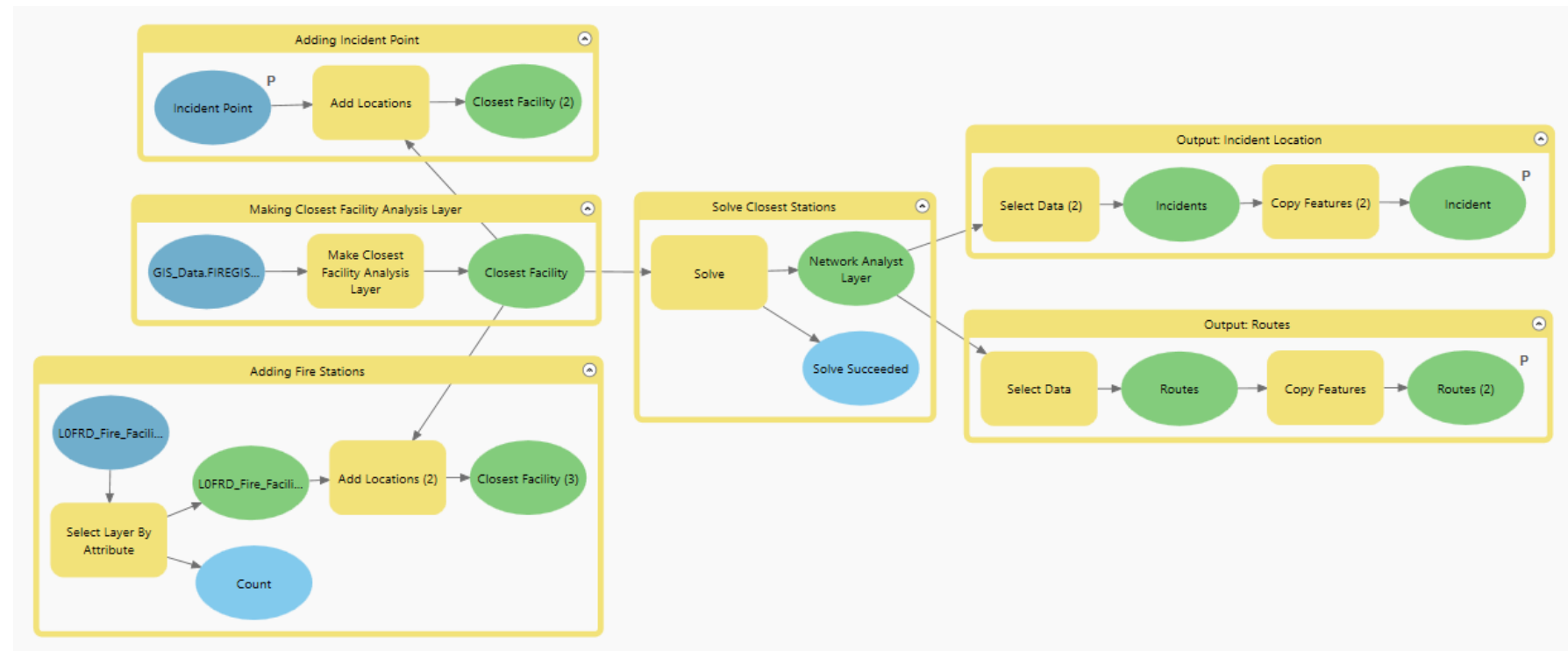
Example: <https://webadaptorhost.domain.com/webadaptorname/rest/services/folder/serviceName/MapServer>

Back Cancel Save

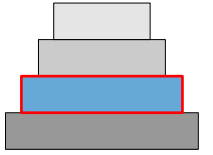
Publishing Custom Geoprocessing Tool: Closest Fire Station



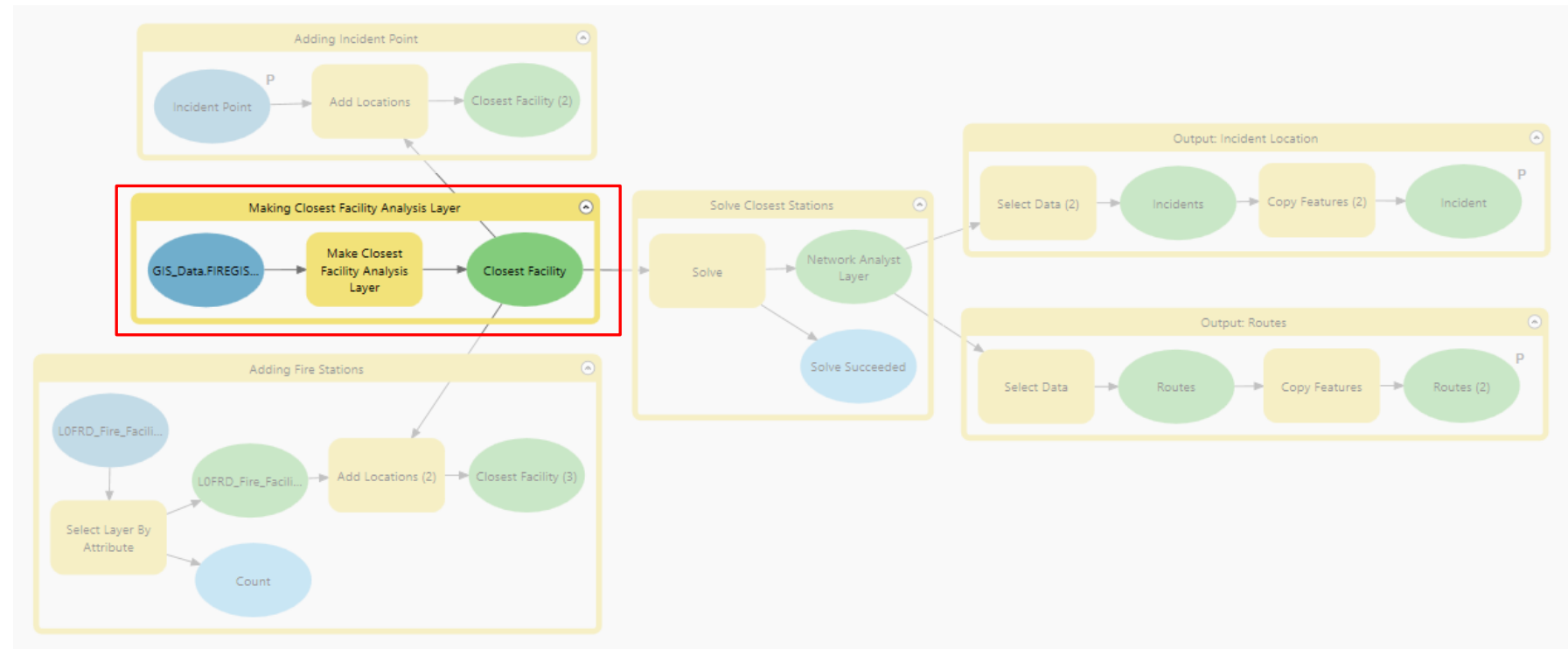
- Make Analysis Layer
- Add Incident Point
- Add Fire Stations
- Solve
- Output
 - Incident Location
 - Routes



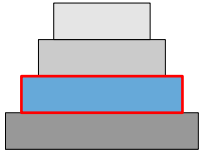
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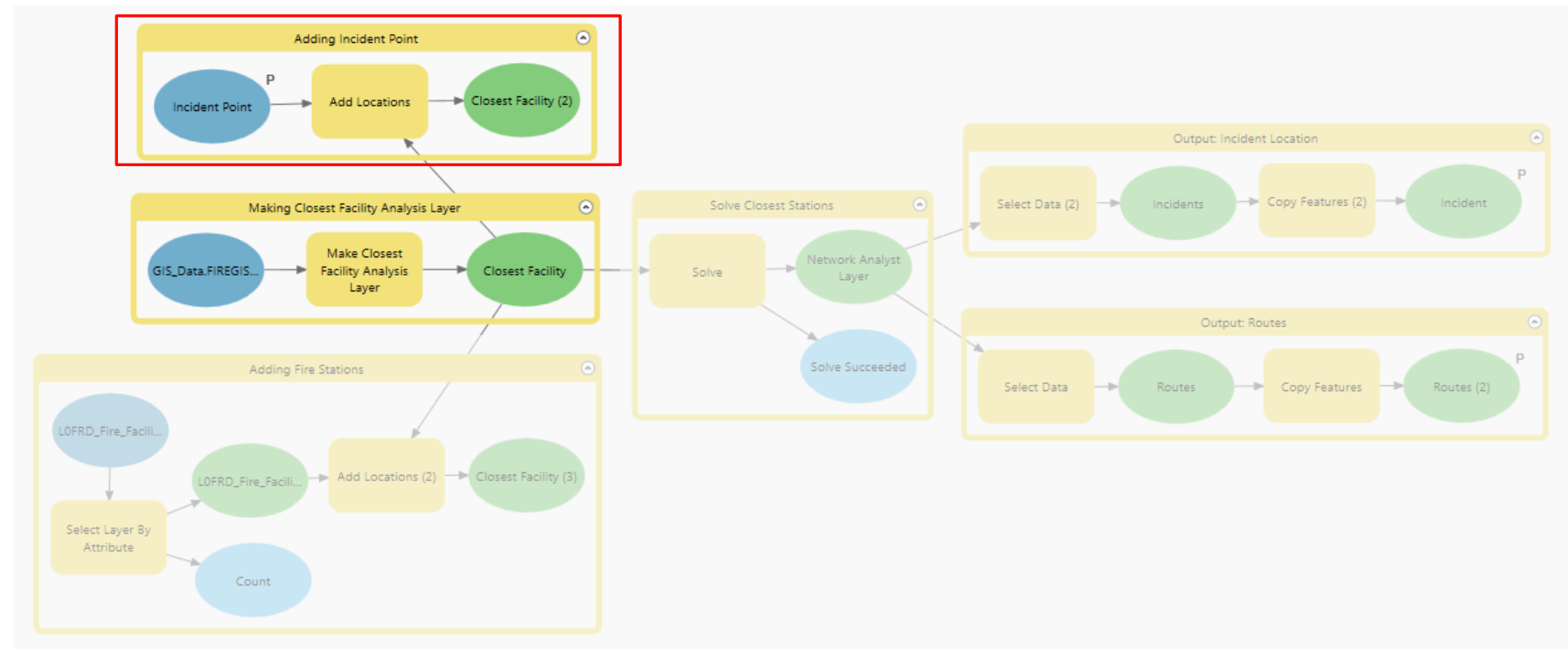
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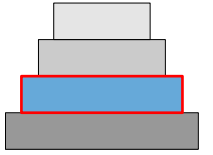
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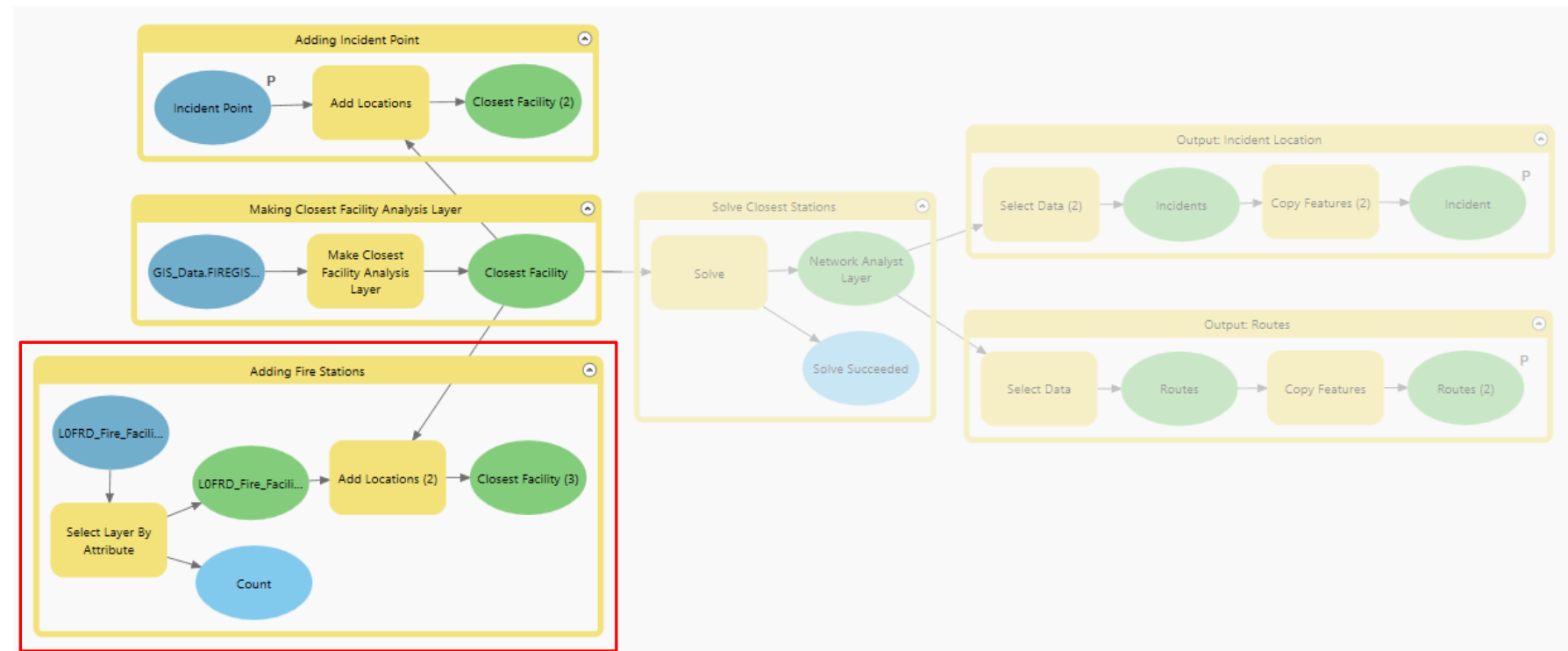
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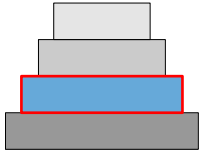
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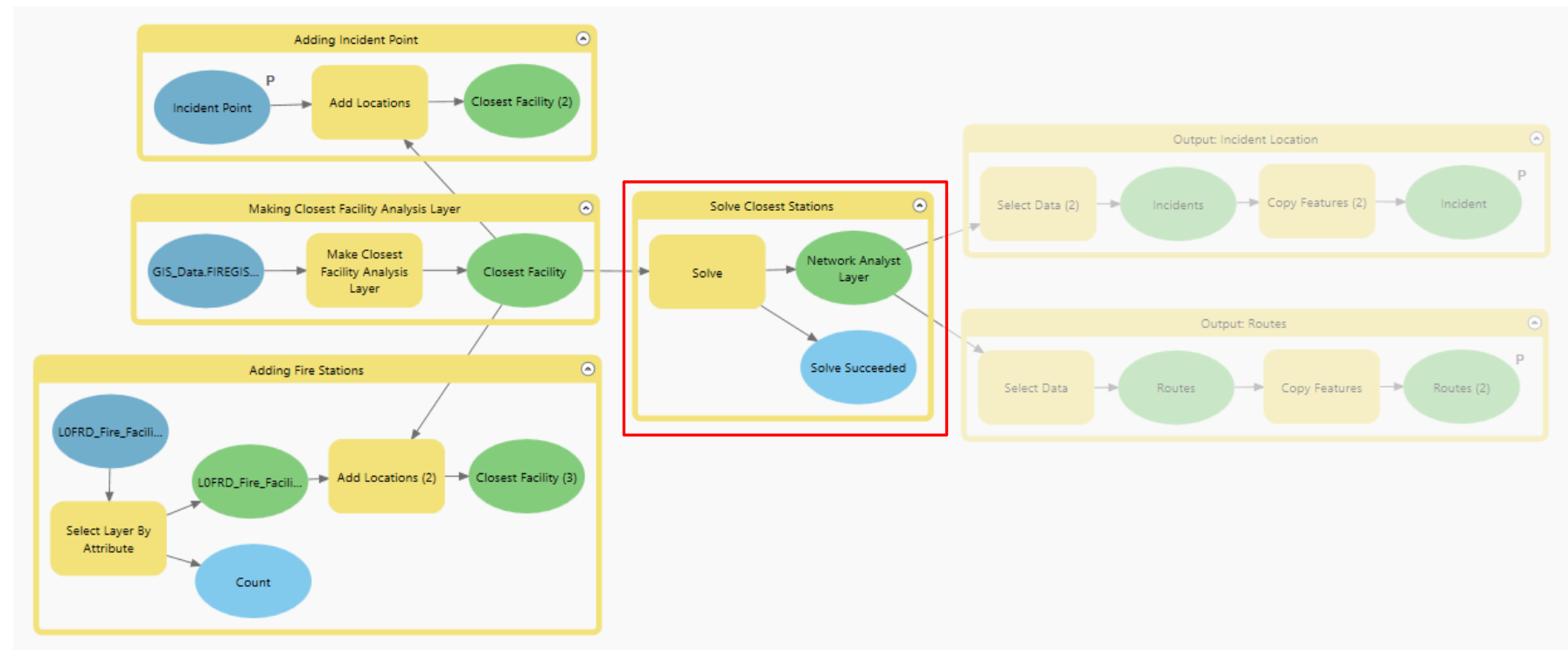
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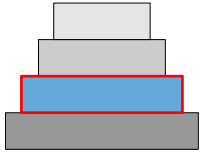
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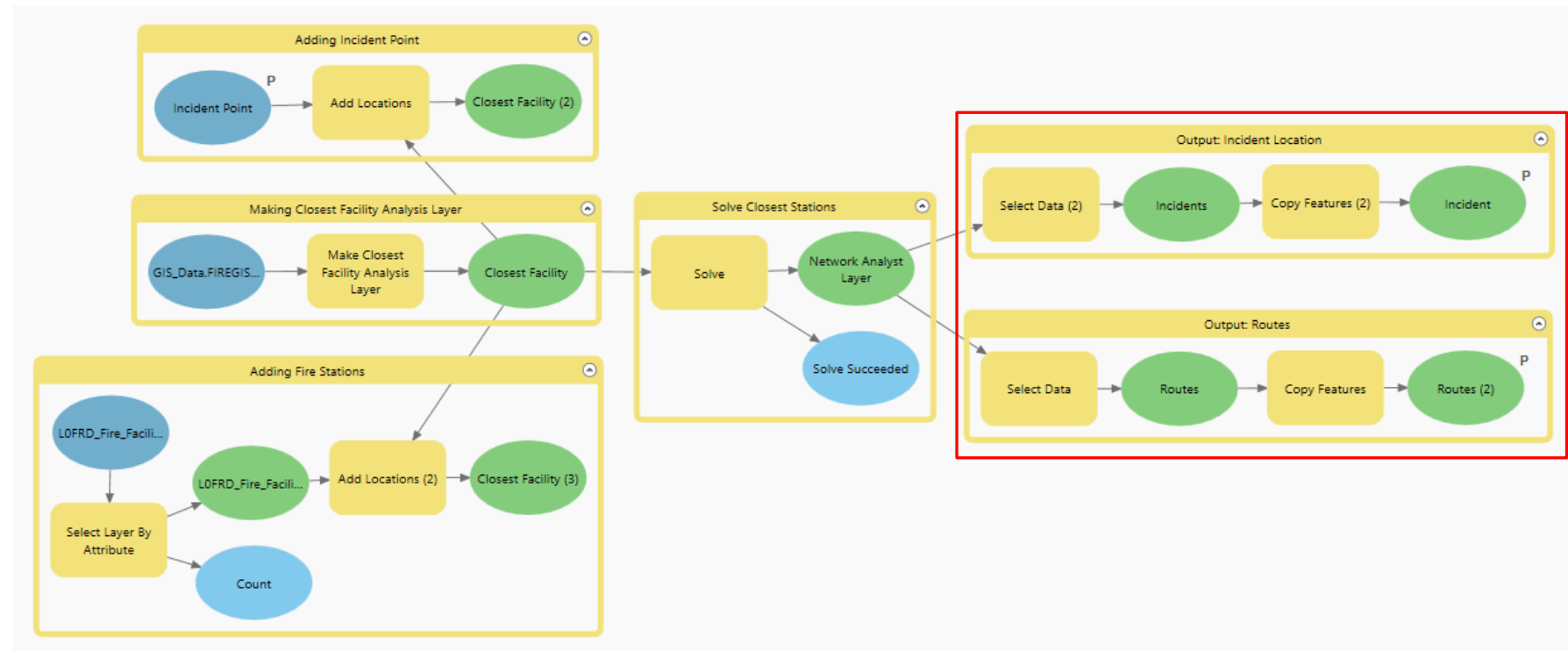
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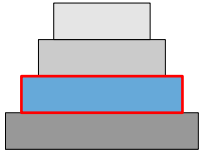
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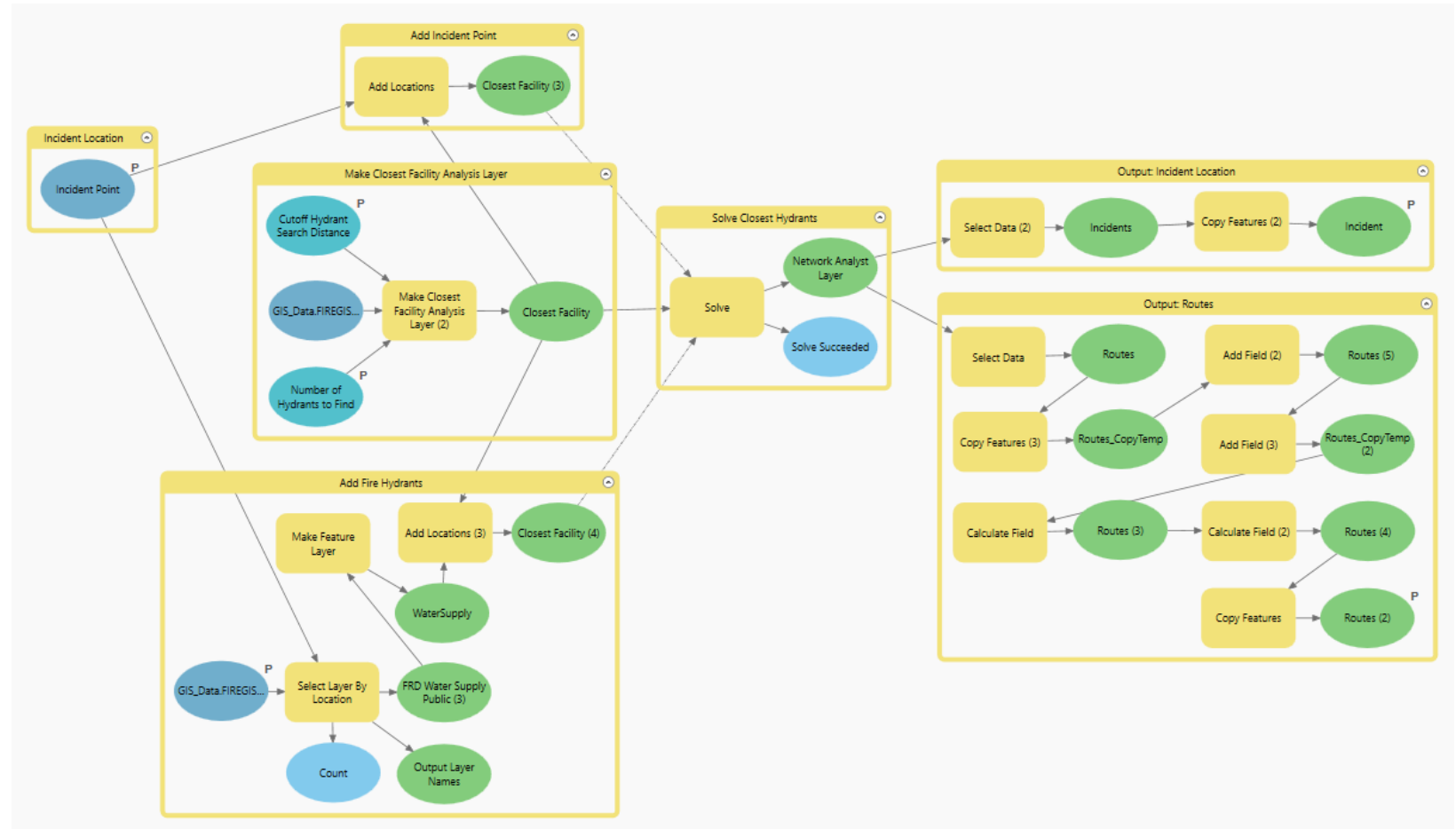
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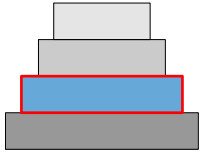
Publishing Custom Geoprocessing Tool: Closest Fire Hydrant



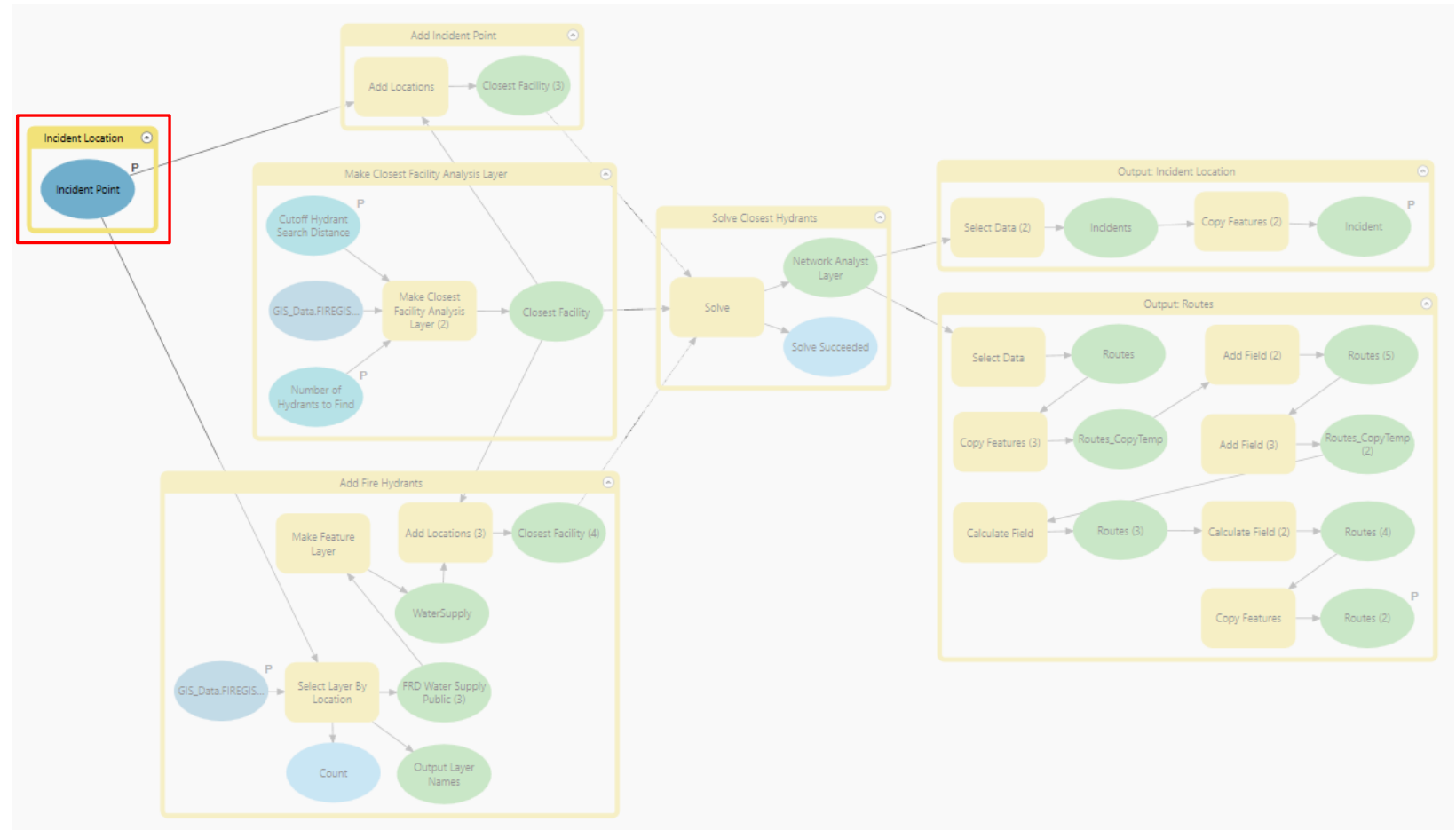
- Incident Location
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 - Routes



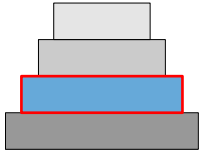
Publishing Custom Geoprocessing Tool: Closest Fire Hydrant



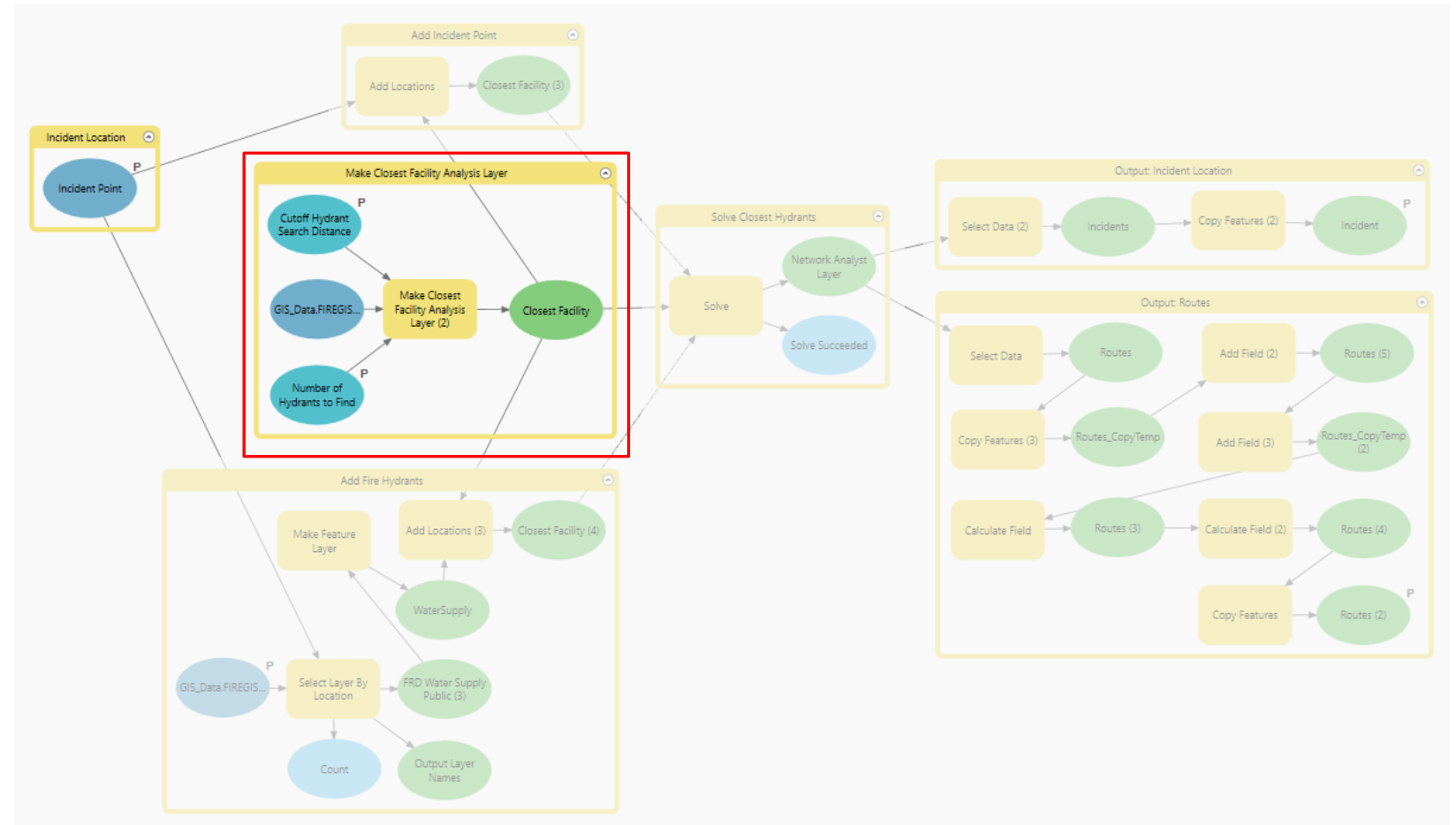
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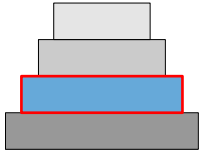
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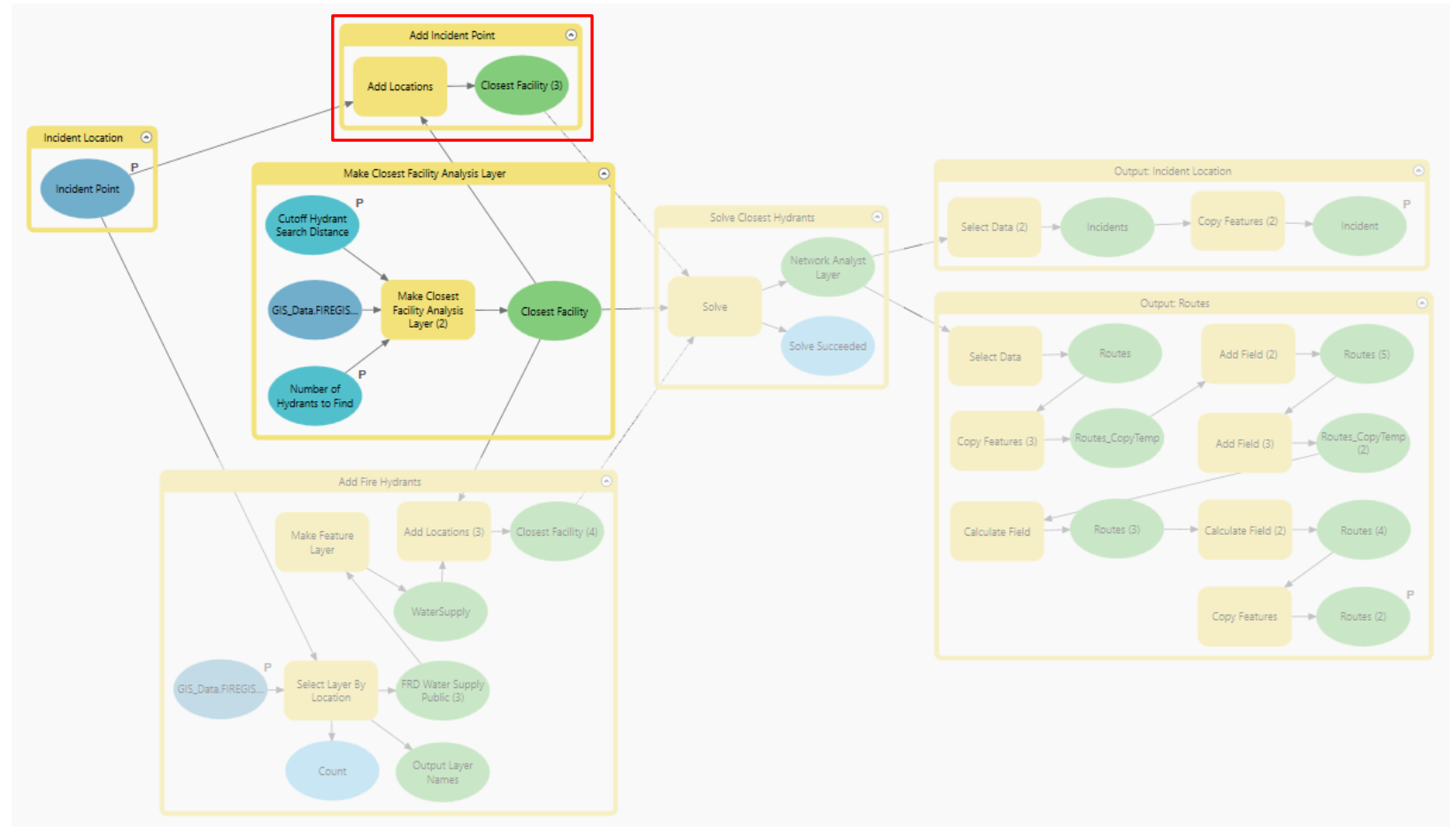
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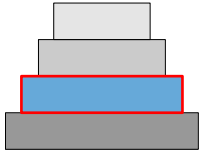
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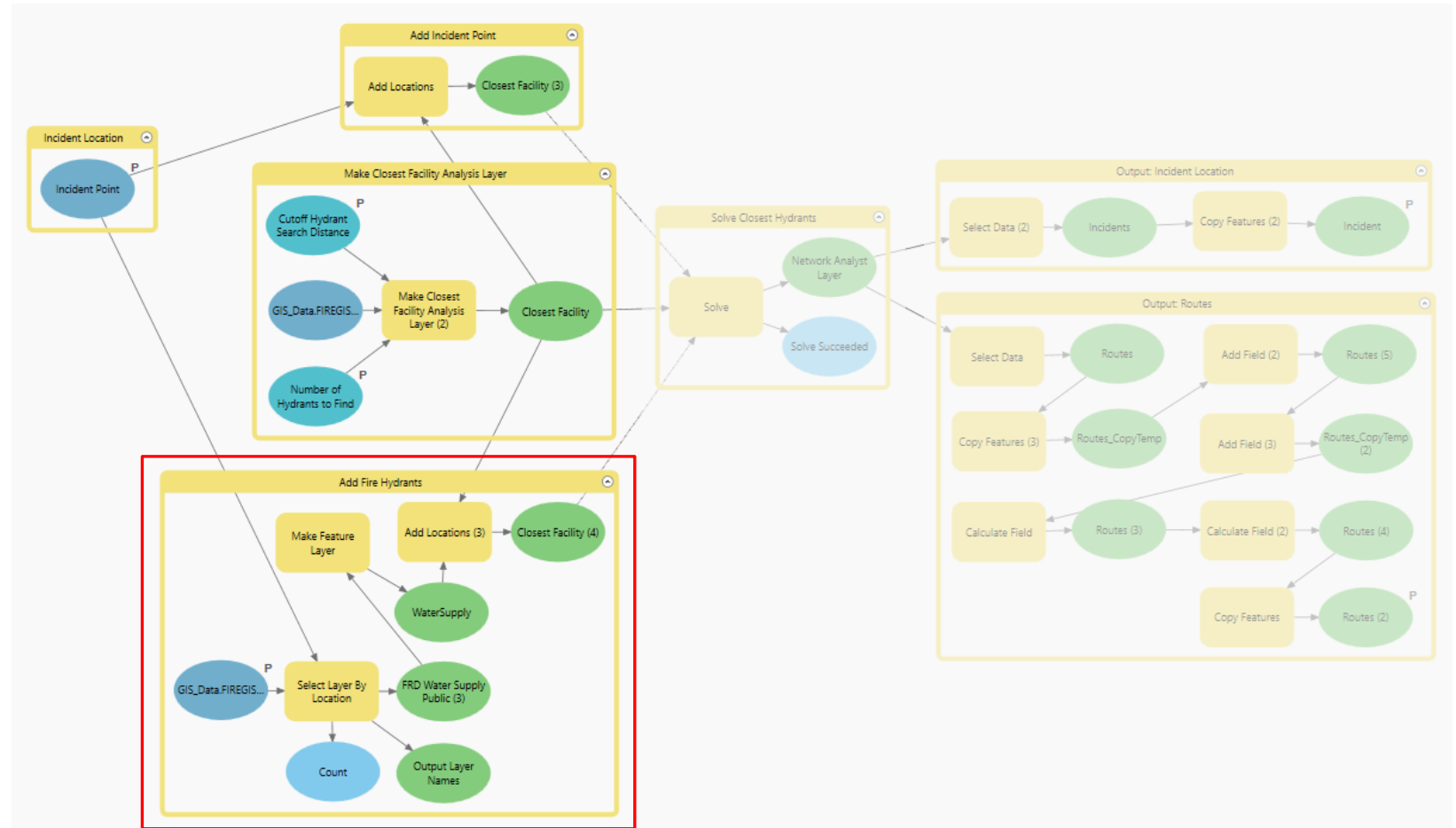
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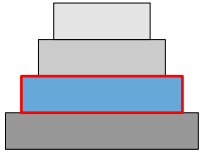
Publishing Custom Geoprocessing Tool: Closest Fire Hydrant



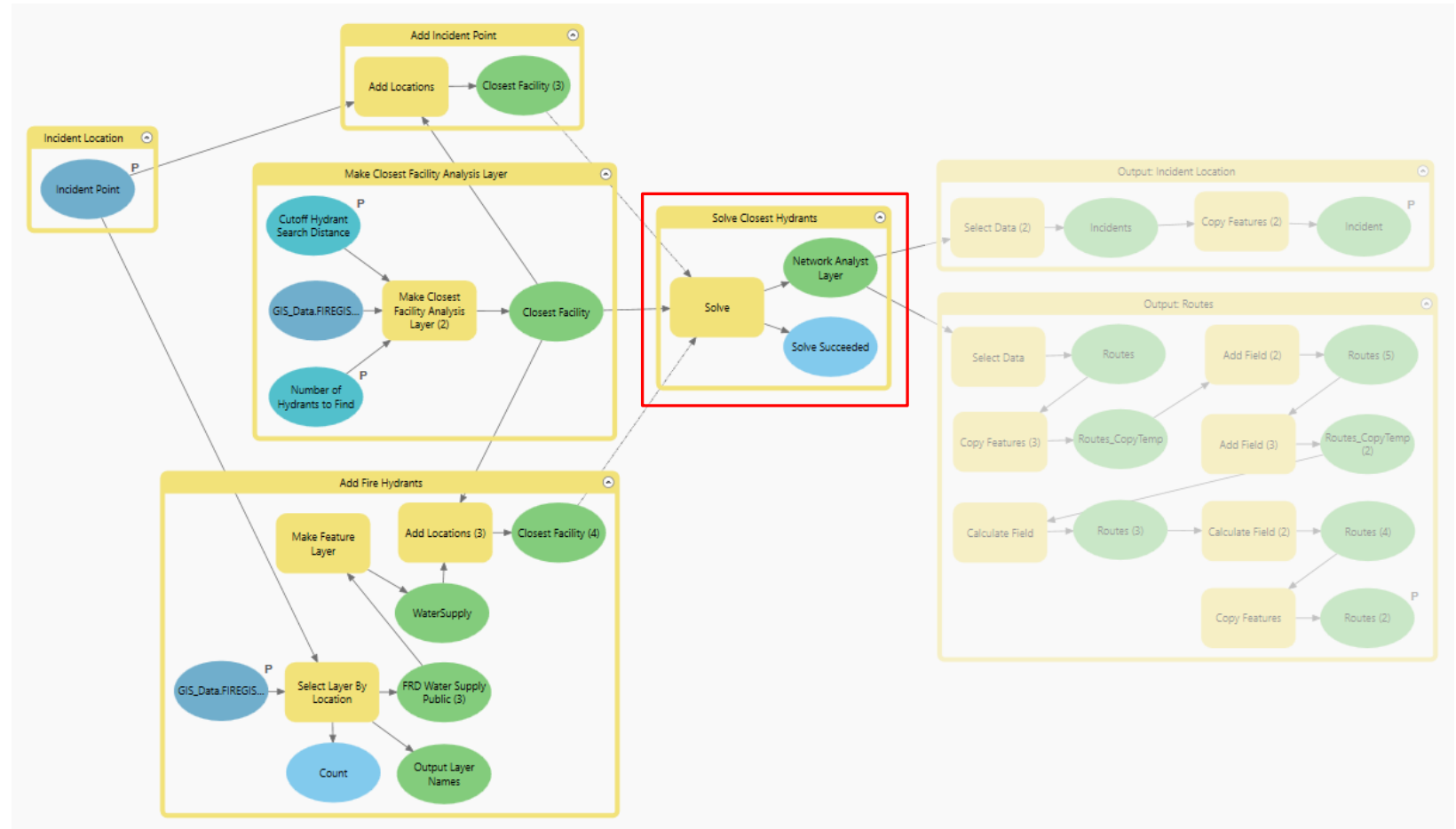
- Incident Location
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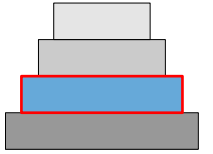
Publishing Custom Geoprocessing Tool: Closest Fire Hydrant



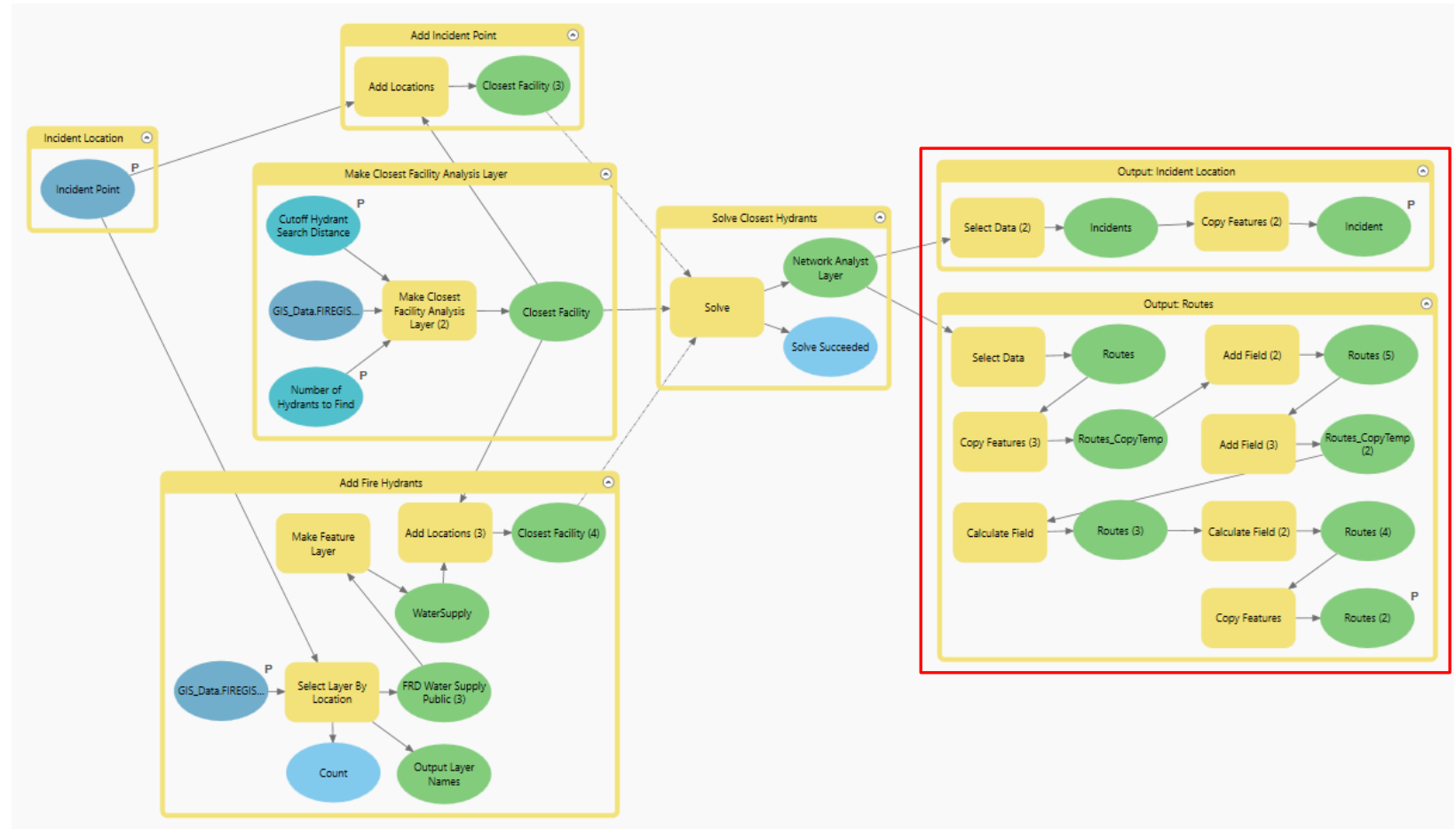
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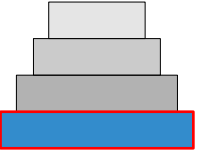
Publishing Custom Geoprocessing Tool: Closest Fire Hydrant



- Incident Location
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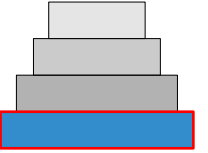


Network Dataset Products Overview



- Static Run Orders
- Fire Box maps with turn-by-turn directions
- Hydrant 2000
- Portal based network dataset
- Portal applications
 - Run Order Visualizer
 - Rescue Router



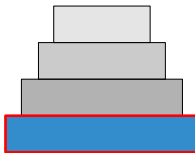


Products – Run Orders

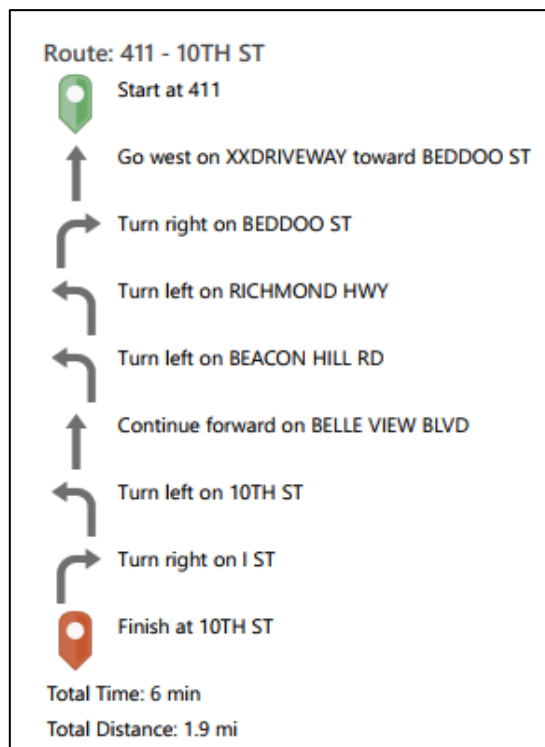
- CAD dispatch and response order are based on nearest available unit.
- Fire Box data is only used in the event CAD is down & incident response plans.
- Static Run Orders
 - Regional Station to Regional Fire Box centroid order of expected arrival
 - Origin – Destination Analysis
 - Pivot table created in Power BI
 - Table provided to CAD staff

Run Orders Report (Fire)																				
Box	Due1	Due2	Due3	Due4	Due5	Due6	Due7	Due8	Due9	Due10	Due11	Due12	Due13	Due14	Due15	Due16	Due17	Due18	Due19	Due20
30301	436	415	404	303	431	425	302	635	624	421	611	615	417	623	438	304	619	439	440	442
30356	303	436	623	415	624	635	304	615	611	404	431	302	425	609	619	421	417	606	438	625
30360	303	624	635	615	611	404	436	623	304	425	302	431	415	606	618	625	609	439	619	622
40100	401	444	429	413	106	108	103	442	710	402	418	428	102	726	430	104	425	110	412	434
40101	401	444	429	413	106	442	418	428	402	108	430	710	103	102	425	434	408	104	410	433
40102	401	444	429	110	108	103	413	106	710	442	104	711	102	402	726	418	105	428	101	430
40103	401	444	106	429	413	108	103	418	428	442	402	102	430	710	104	434	410	425	110	101
40105	401	444	429	710	413	106	442	726	418	711	402	430	720	412	108	434	425	428	408	730
40107	401	444	106	429	413	418	428	108	442	103	402	430	102	434	710	410	104	433	425	110
40108	401	106	444	413	429	418	428	108	430	103	434	102	402	442	433	104	410	408	423	110
40109	401	444	429	710	413	442	412	106	726	711	418	402	430	720	434	108	425	428	730	741
40110	401	444	429	710	413	442	106	726	412	711	402	418	430	720	108	434	425	428	730	741
40111	401	444	429	106	413	418	428	442	402	108	430	103	710	102	434	425	410	104	408	433
40112	401	444	429	710	413	106	442	726	108	418	711	402	430	720	103	412	434	425	428	408
40116	401	444	710	429	110	413	726	108	103	711	106	442	720	104	418	402	730	741	430	102
40141	401	444	108	429	103	106	413	102	442	710	428	104	402	110	418	101	726	430	425	412
40142	108	401	103	106	102	104	110	444	428	101	418	429	710	711	413	109	410	726	442	105
40143	401	106	108	444	103	413	428	429	102	418	104	110	101	430	442	410	710	434	402	433
40144	401	108	103	106	444	102	428	104	429	110	413	101	418	442	710	410	109	430	711	402
40146	401	444	429	710	413	110	108	103	106	442	726	402	104	711	418	102	428	430	720	412
40147	401	444	429	108	413	106	103	442	710	102	402	418	428	104	110	726	430	101	425	412
40148	401	444	710	429	110	413	726	108	103	711	106	442	720	104	418	402	730	741	430	102
40149	401	444	108	103	110	429	710	106	711	104	102	413	105	726	101	442	428	402	418	720
40162	401	110	444	108	103	429	104	710	106	711	413	102	105	101	726	442	428	402	209	109
40163	401	444	710	429	413	110	726	108	103	711	106	442	720	104	418	402	730	741	430	102
40164	401	444	710	110	108	103	429	413	726	104	106	711	442	102	720	105	101	418	402	730
40165	401	444	710	429	413	110	726	108	103	711	106	442	720	104	418	402	730	741	430	102
40166	401	444	710	429	110	413	726	108	103	711	106	442	720	104	418	402	730	741	430	102
40167	401	444	110	108	103	710	429	106	104	711	413	102	726	105	101	442	428	720	402	418
40168	401	444	710	429	413	726	106	711	442	720	418	402	430	730	412	741	706	434	108	425
40169	401	444	110	108	103	710	429	106	104	711	413	102	726	105	442	101	428	402	720	418
40170	401	444	710	429	413	726	106	711	442	720	418	402	430	730	412	741	706	434	108	425
40190	106	108	401	103	428	102	418	104	444	110	101	413	410	429	430	109	434	206	408	402
40191	106	401	108	428	444	413	418	103	102	429	104	430	410	110	101	434	402	433	408	423
40192	106	401	444	413	418	428	108	429	430	102	103	434	410	104	402	433	442	110	408	101
40193	401	106	444	413	108	428	429	418	103	102	430	104	442	434	410	110	101	402	433	408

Products – Fire Box Maps



■ Fire Box Maps



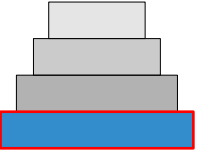
Turn by turn directions



Color fire box maps

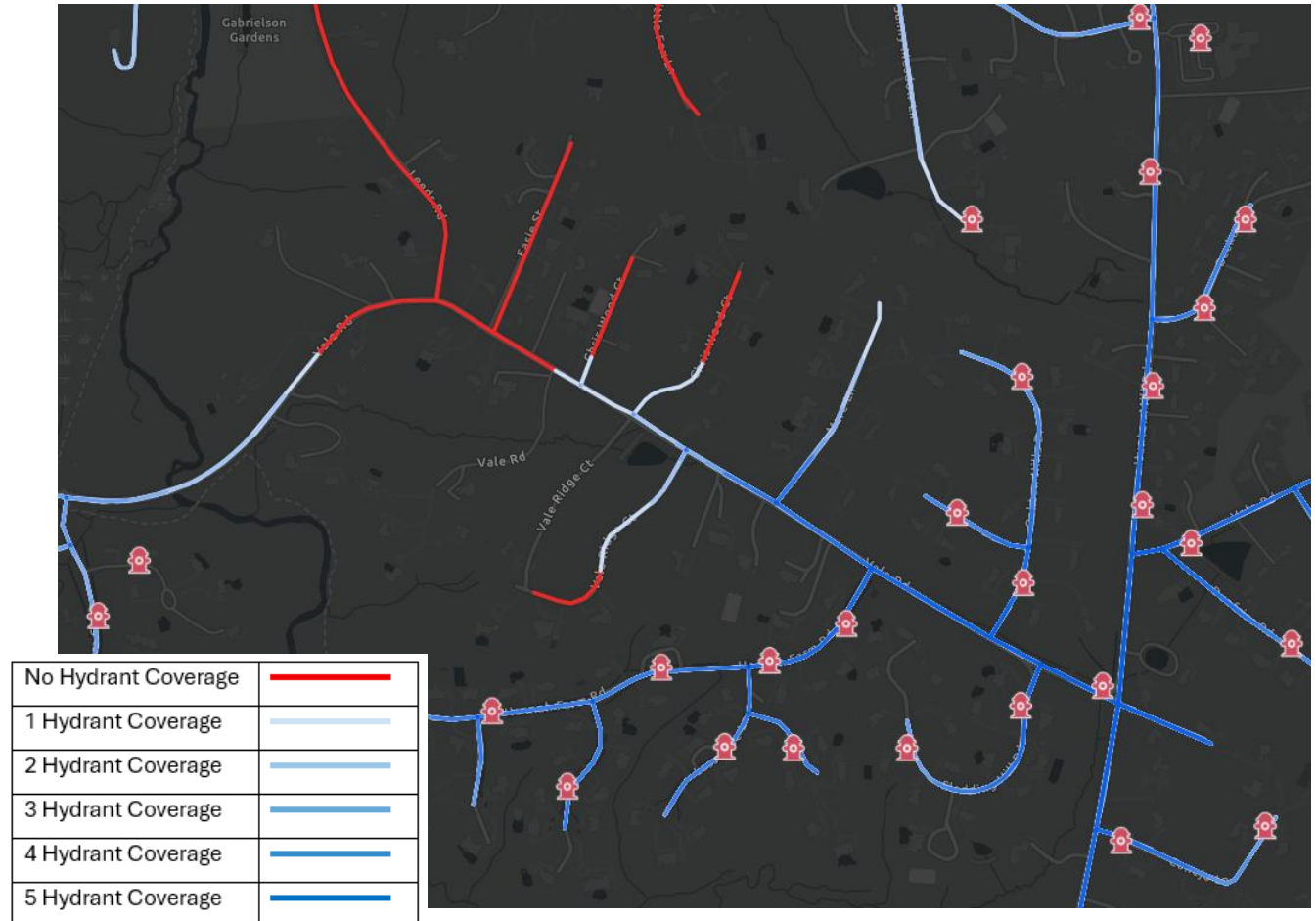
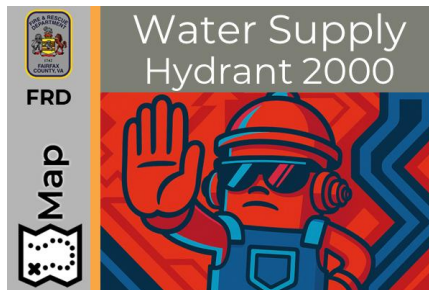


Black and white fire box maps

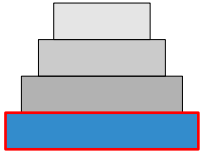


Products – Hydrant 2000

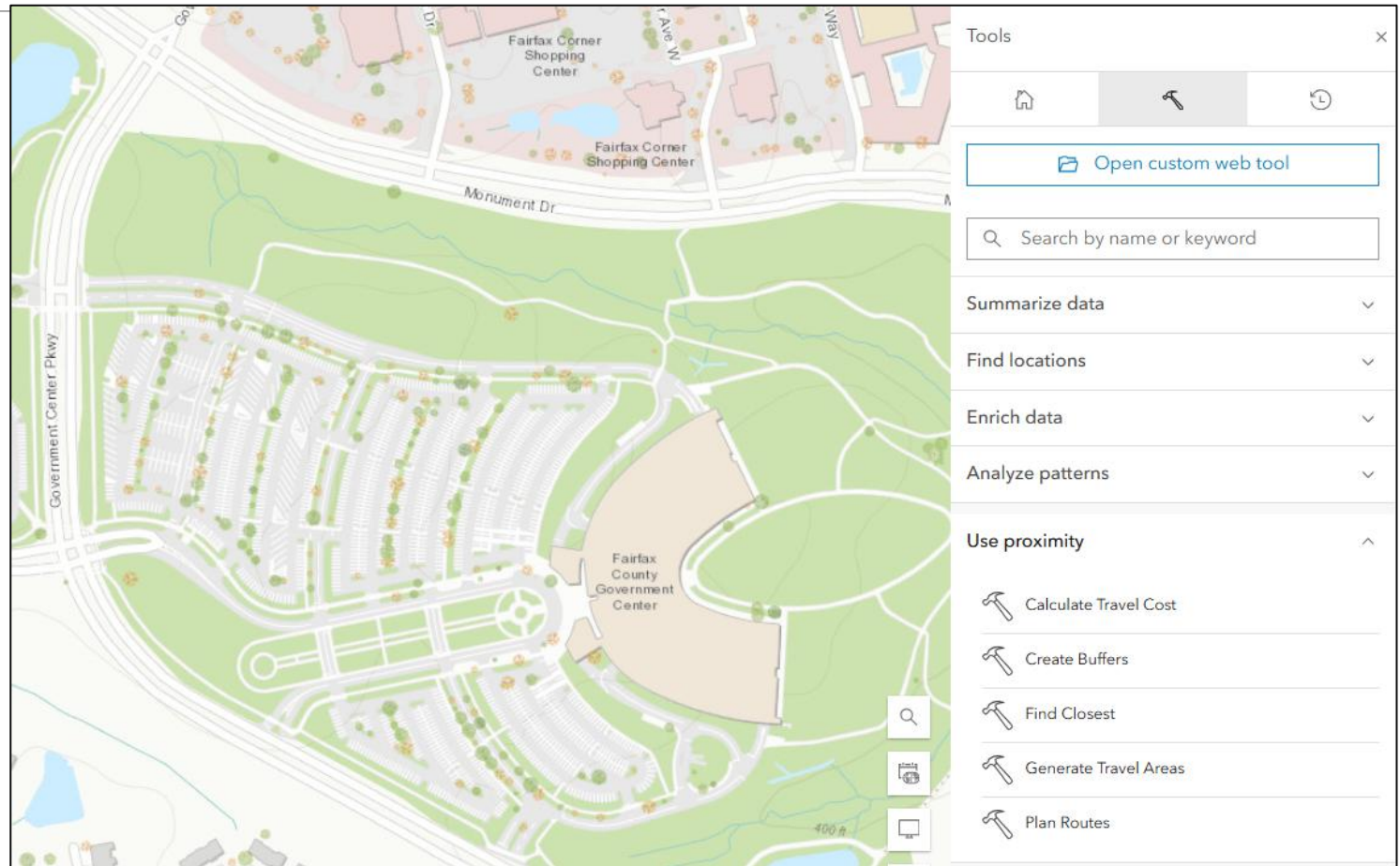
- Service Area one time analysis
 - 2000 feet from every hydrant
- Dark blue lines represent hydrant coverage
- Red lines indicate no hydrant coverage within 2000 feet
- Power Automate used to change operational status of hydrants



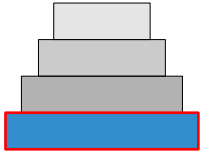
Products – Portal Based Analysis Tools



- Network Analysis Routing Service
- Hosted internally
- Supports common workflows
 - Route
 - Closest Facility
 - Service Area
 - Origin-Destination Cost Matrix
- Provides consistent routing results
- No credit usage



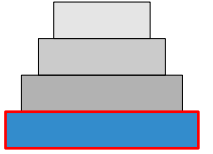
Network Dataset Applications



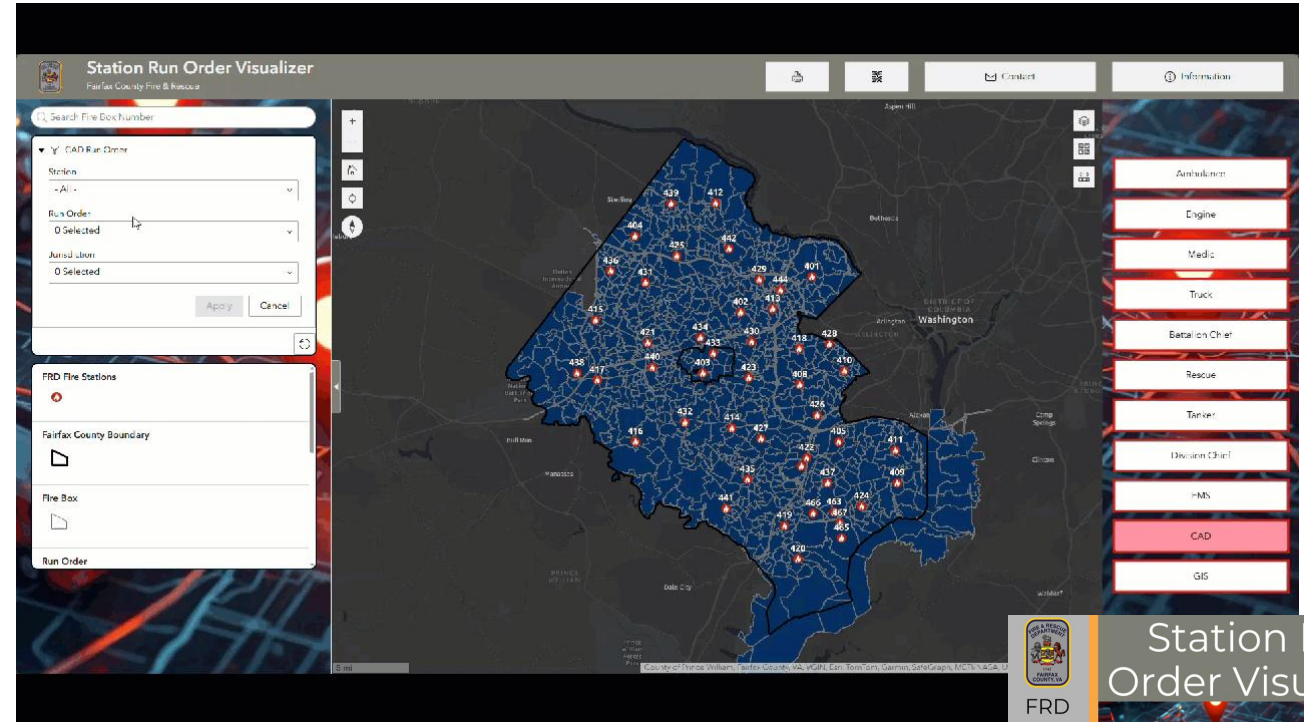
- Station Run Order Visualizer
- Rescue Router (routing app)
- Fire Station Locator – Stations & Hydrants



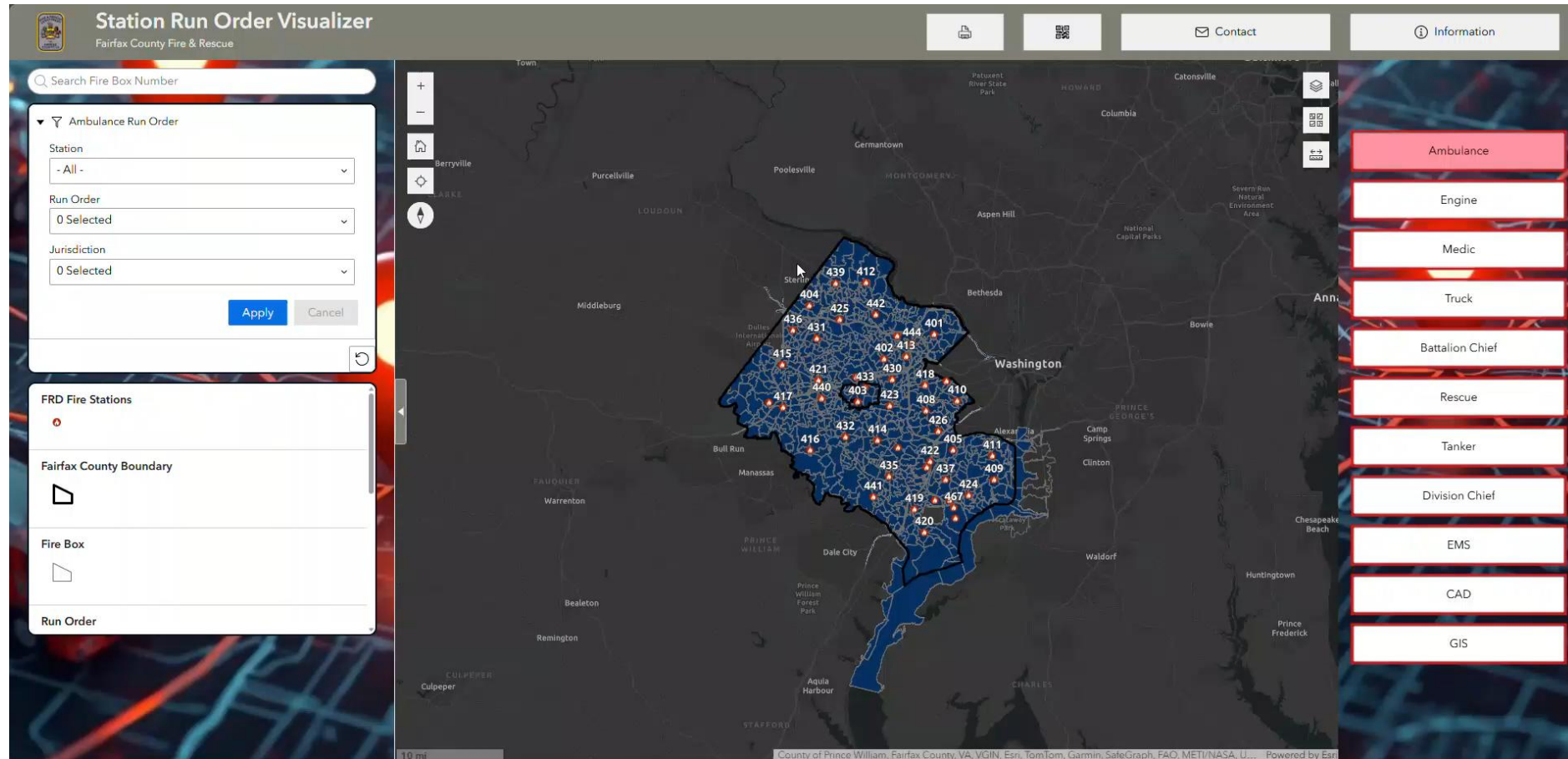
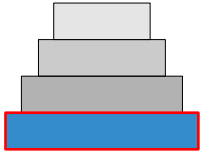
Applications – Station Run Order Visualizer



- Purpose
 - Visualize CAD and GIS run order data by fire box
 - Identify errors
- Data/Layers
 - Joined layer view
 - Run order table + fire box polygons (authoritative)
- Filters
 - Apparatus type
 - Station
 - Run order

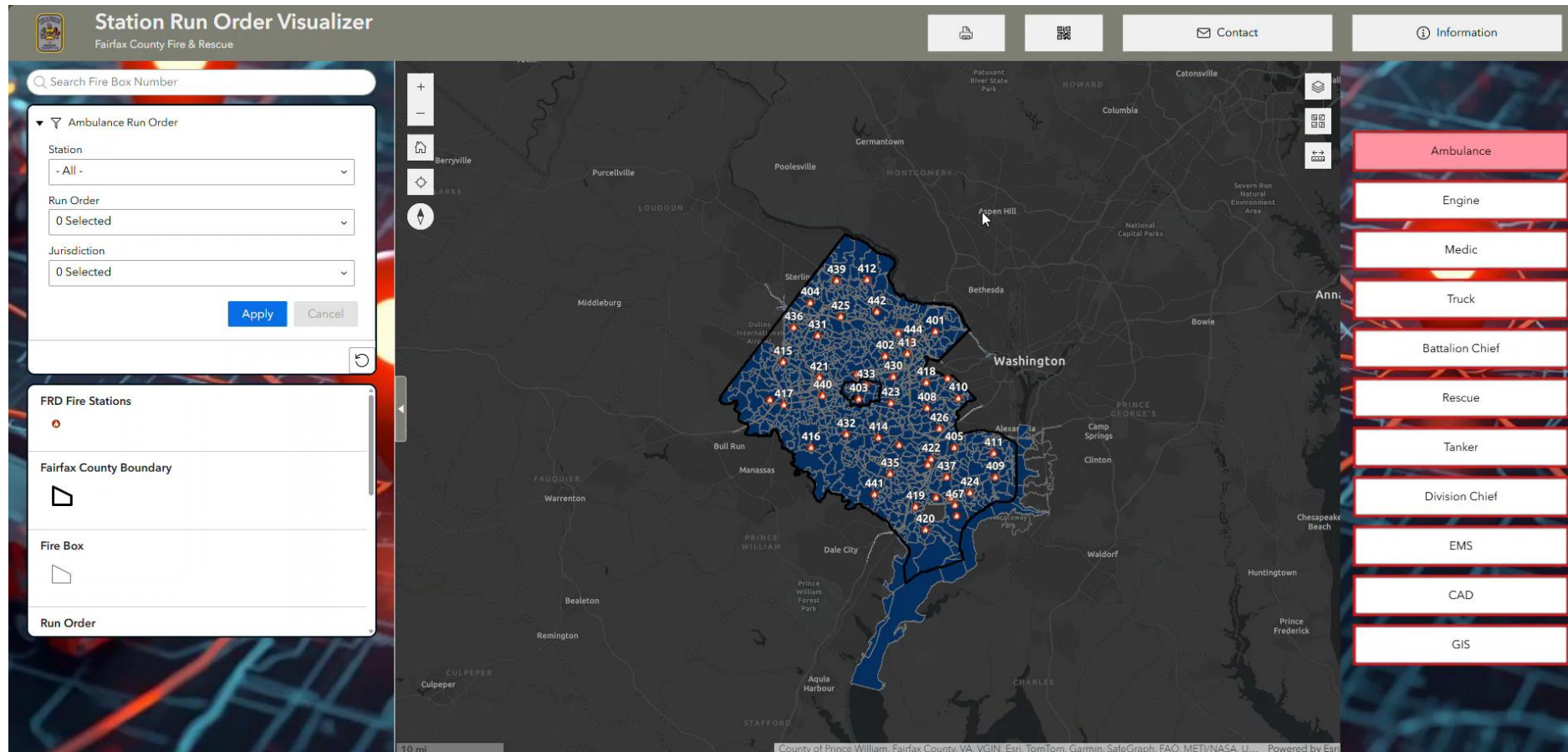
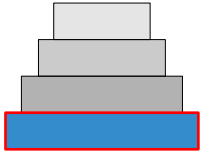


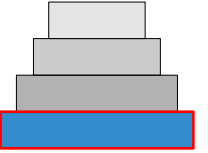
Applications – Station Run Order Visualizer: Data Discrepancies



Applications – Station Run Order Visualizer

Filtering by Apparatus

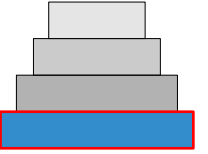




Applications – Rescue Router

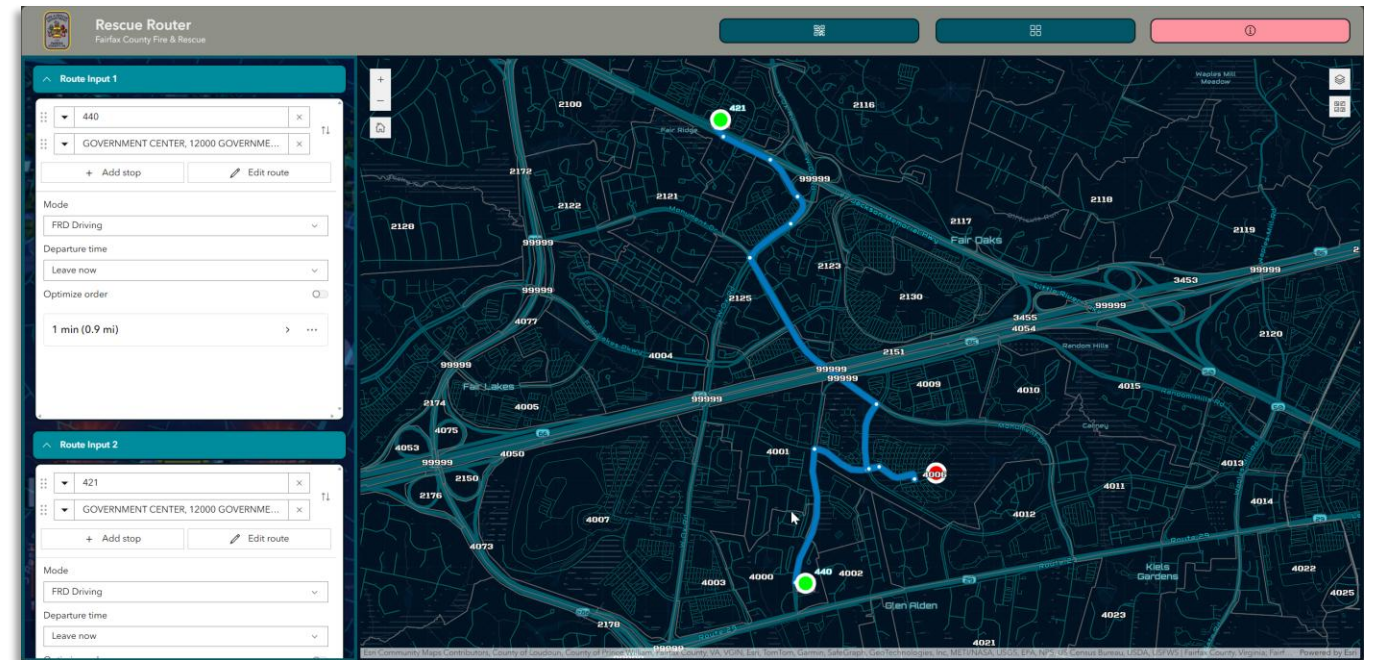
- Purpose
 - View & compare routing analysis
 - Identify data discrepancies/errors and verify routes
 - Mimics CAD closely but not exactly
 - Differences based on network dataset configurations
 - Improves efficiency
- 3 analysis tools
 - Apparatus Routing
 - Closest Stations
 - Closest Hydrants
- Automatic Vehicle Location (AVL)
 - Future enhancement

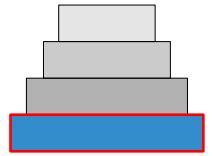




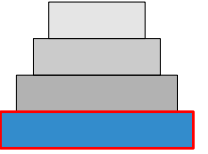
Applications – Rescue Router

- Apparatus Routing
 - Directions widget
 - Using FRD Network Dataset
 - View & verify routes
 - Users can choose between travel modes
 - 2 inputs to compare route details
 - Compare routes to incidents in areas of “competitive fringe”
 - Can lead to improved data quality
 - Locations are only addresses or fire stations



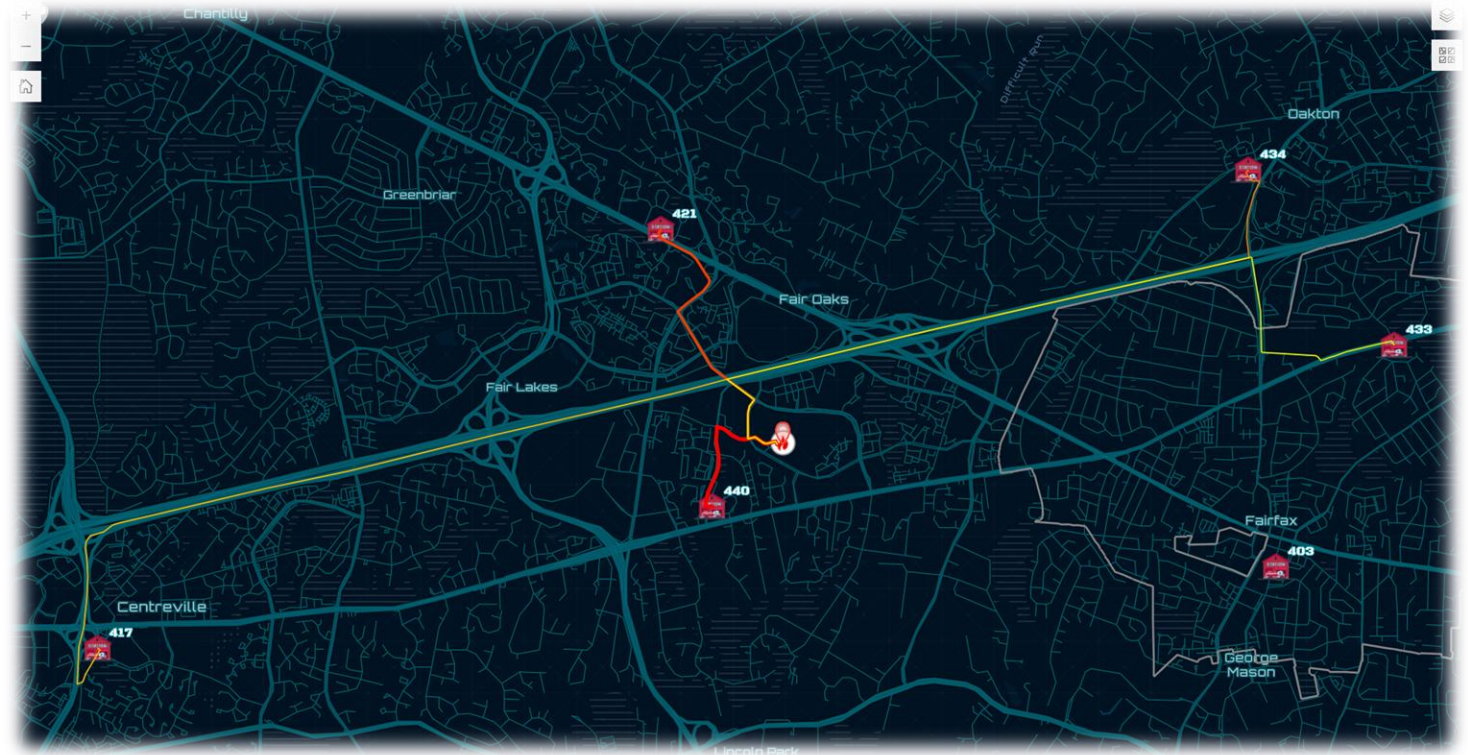


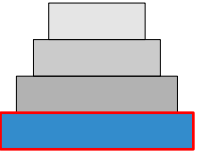
Applications – Rescue Router



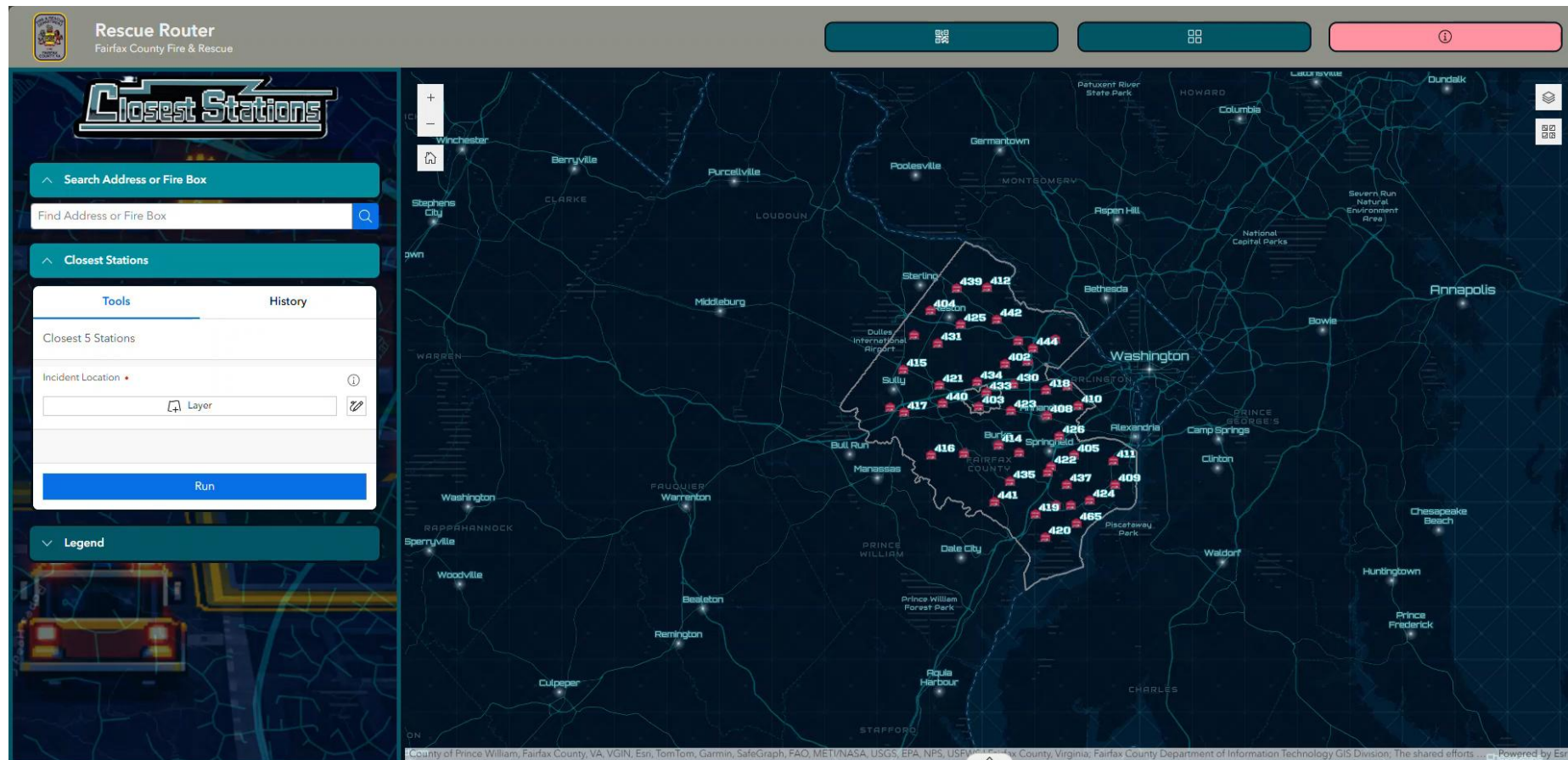
Applications – Rescue Router

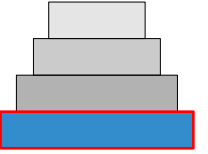
- Closest stations to an incident
 - Custom analysis tool
 - User input does not need to be an address
 - How it works:
 - Drop a point on the map
 - Analysis tool runs
 - Output = closest 5 stations by drive time





Applications – Rescue Router

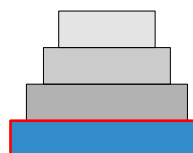




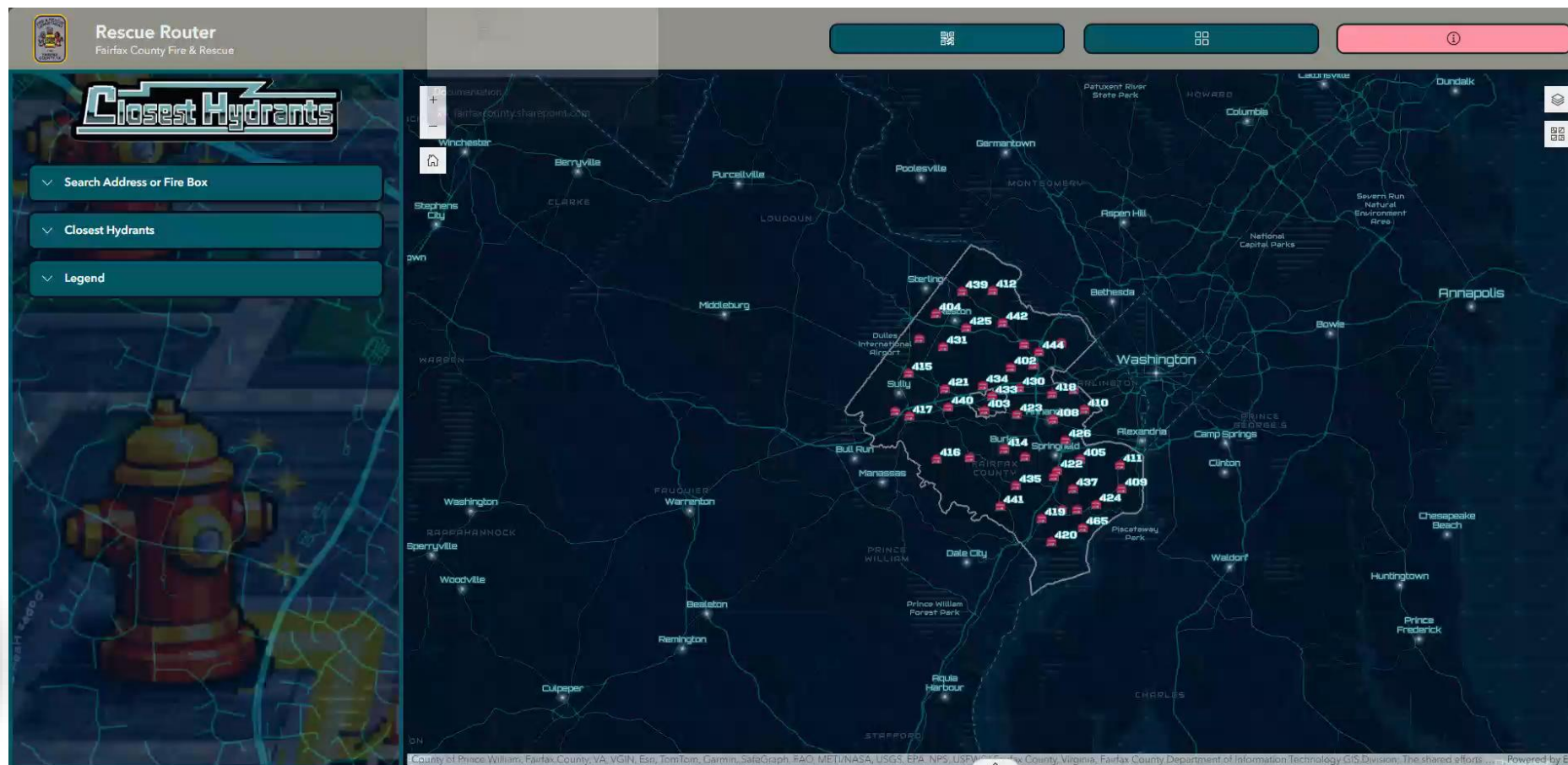
Applications – Rescue Router

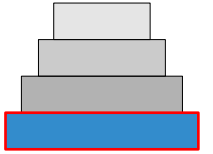
- Closest hydrants to an incident
 - Custom analysis tool
 - User input does not need to be an address
 - User also has control over search distance and output feature amount
 - How it works:
 - Drop a point on the map
 - Analysis tool runs
 - Output = defaults to 5 closest hydrants within 2000 ft of user input





Applications – Rescue Router

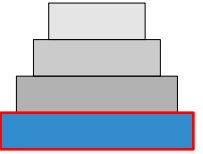




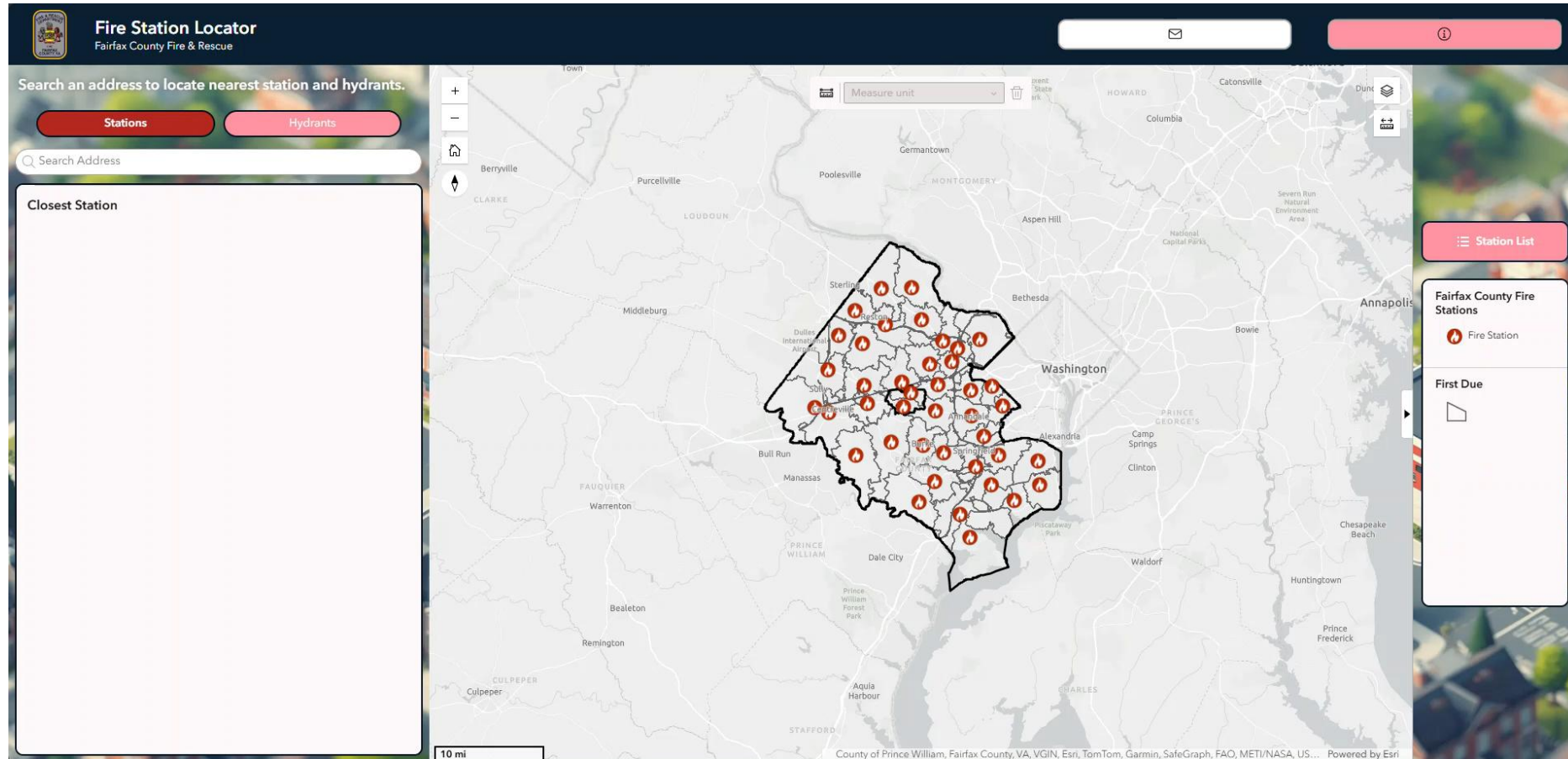
Applications – Public Facing

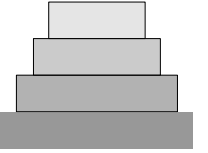
- For public facing apps, we have opted to use 3rd party routing networks
- Purpose
 - Users can identify the fire station that responds to their address and locate closest hydrants
 - Does NOT mimic CAD routing, **not used for emergency routing**
- Third Party Network
 - Custom pop-up configuration that links to Google and/or Apple maps
 - Users are more familiar with these interfaces





Applications – Public Facing





Future Enhancements

- Schema refinement – for apparatus specific travel modes
 - Clearance (Height)
 - Weight
 - Width
- Convert Model Builder to Python scripts
 - More reliable & ability for more detailed tool messages
- Indoors
 - Multi-modal routing recommendations
- AGOL apps
 - Public Routing
- Portal apps
 - Incident Run Orders

Event Type	Condition 1	Condition 2	Condition 3
FGAPT (Fire: Garden apartment)	4E, 2(T or TL or TT), 1(R or HM401), 1(A or M), 1M, 1(ALS or EMS), 2BC, 1(SAF or SAF403) *1COUNTY BC Required		
FHIRIS (Fire: High rise)	5E, 2(T or TL or TT), 1(R or HM401), 1(A or M), 1M, 1(ALS or EMS), 2BC, 1(SAF or SAF403) *1COUNTY BC Required		
FHOSP (Fire: Hospital)	5E, 2(T or TL or TT), 1(R or HM401), 1(A or M), 1M, 1(ALS or EMS), 2BC, 1(SAF or SAF403), 1CSU *1COUNTY BC Required		
FHOU (Fire: House)	4E, 2(T or TL or TT), 1(R or HM401), 1(A or M), 1M, 1(ALS or EMS), 2BC, 1(SAF or SAF403) *1COUNTY BC Required *Rural Water Includes 2K		
FJAILF (Fire: Jail)	5E, 2(T or TL or TT), 1(R or HM401), 1(A or M), 1M, 1(ALS or EMS), 2BC, 1(SAF or SAF403) *1CITY BC Required		
FLSLG (Flam liquid spill large > 55 gal)	4E, 1(T or TL or TT), 1HM401, 1HMS401, 1R(Hazmat), 1R, 2(A or M), 1BC, 1(ALS or EMS), 1SAF -FOAM TASKFORCE		
FLSSM (Fire: Flam liquid spill small < 55 gal)	1E, 1(R-hazmat or 1HM401) *1HMS401 Required, If HM401 Recommended		
FMADMIN (FM Admin)	1IV (Self dispatch)		
FMCOURT (FM Court)	1IV (Self dispatch)		
FMETROF (Fire: On metro property - No train)	2E, 1(R or HM401), 1(T or TL or TT), 1BC, 1(SAF or SAF403)		
FMFOLLOW (FM Follow)	1IV (Self dispatch)		
FMIDRIS (Fire: Mid rise)	5E, 2(T or TL or TT), 1(R or HM401), 1(A or M), 1M, 1(ALS or EMS), 2BC, 1(SAF or SAF403) *1COUNTY BC Required		



Questions?

Contact Info:

Data Analytics Strategy Management
Fairfax County Fire and Rescue Department

www.fairfaxcounty.gov/fire-ems

charles.galloway@fairfaxcounty.gov

john.woolvertton@fairfaxcounty.gov

fire.GIS@fairfaxcounty.gov

